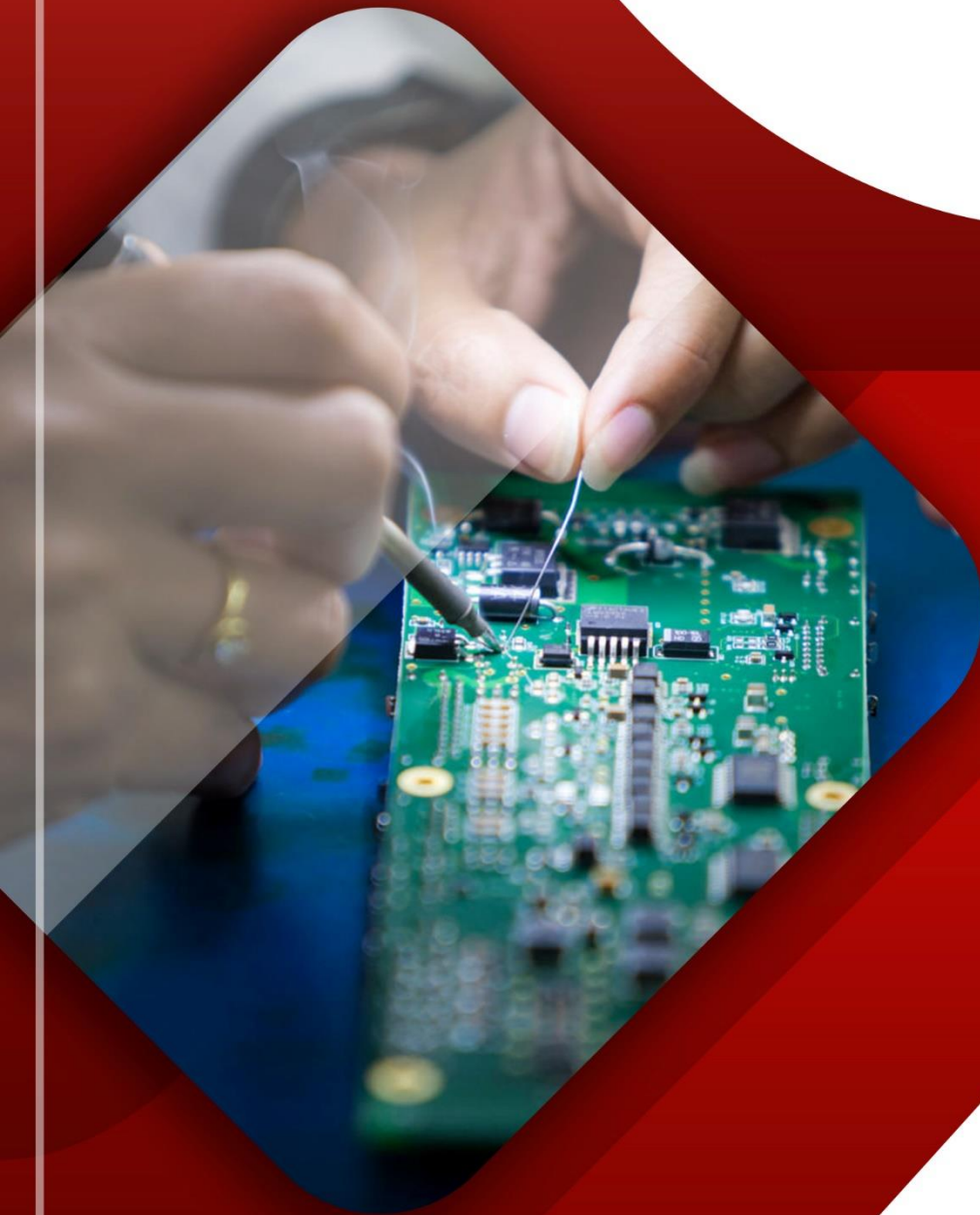


LABORATORY MANUAL IN **CIRCUIT 2**



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OSCILLOSCOPE OPERATION

Experiment # 1

Introduction

The experiment on Oscilloscope Operation is designed to introduce students and enthusiasts to the fundamental working principles and applications of an oscilloscope, a critical instrument in electronic laboratories and various engineering fields. An oscilloscope, often referred to as an 'oscope', is a type of electronic test instrument that graphically displays varying signal voltages, usually as a two-dimensional plot of one or more signals as a function of time.

This hands-on experiment aims to familiarize participants with the basic controls, functions, and methods of measurement that are possible with an oscilloscope. Key learning objectives include understanding the oscilloscope's interface, learning how to set up the device for different types of signals, and interpreting the waveforms displayed.

Participants will engage in practical activities, starting with the setup of the oscilloscope, including connecting probes and calibrating the instrument. They will then explore the different modes of operation, such as AC/DC coupling, adjusting time base and amplitude settings, and using triggering features to stabilize waveforms.

Throughout the experiment, emphasis is placed on safety precautions, especially considering the high voltages often involved in oscilloscope measurements. This foundational knowledge is not only critical for mastering the oscilloscope but also serves as a stepping stone towards more complex electronic diagnostics and design tasks.

By the end of the experiment, participants should be able to confidently operate an oscilloscope, interpret basic waveforms, and understand the significance of various settings and adjustments. This skill set is indispensable for anyone pursuing a career in electronics, electrical engineering, telecommunications, or related fields.

OBJECTIVES

1. To identify the operating controls of an oscilloscope
2. To set up the oscilloscope and adjust the controls properly to observe an ac voltage waveform.

MATERIALS REQUIRED

Power Supply:

- Power source for oscilloscope
- 0-15 V dc variable supply

Instruments:

- Oscilloscope
- DMM

Miscellaneous:

- Assorted leads for connecting to the oscilloscope terminals and jacks

PROCEDURE

1. Power to the oscilloscope should be turned **off**. Examine the front panel of the oscilloscope carefully, noting the types and functions of each control and jack. Examine the back of the scope for additional jacks, switches, and controls.
2. Tabulate the information on the scope on Table 1-1.
3. Turn the oscilloscope **on**. Set the time-base switch to the EXT or X-Y position. If you are using a dual trace oscilloscope, set the control for channel 1 (or A) operation. After a brief warm-up period, a bright spot should appear on the screen. If there is no spot, adjust the intensity control clockwise.
CAUTION: *Do not set the intensity too high and do not leave the spot on too long; the screen coating can be permanently damaged.*
 If the spot still does not appear, it may be off screen. Use the vertical and horizontal positioning controls until the spot becomes visible on the screen.
4. If you already have done so, adjust the applicable controls to produce a sharp, clear, centered spot on the screen. This will involve the vertical and horizontal positioning, focus, astigmatism (use only if an adjustable control is on the front panel), and intensity controls. Note that the focus and intensity controls (and the astigmatism control, if adjustable) interact with one another. Therefore, it is necessary to adjust both (or all three) to obtain the sharpest, thinnest point of light.
5. Set the time-base switch to automatic operation (this may involve simply disengaging the X-Y switch). Adjust the time-base (Time/Div.) control to 1 ms/div and the trigger level control to Auto and positive slope. A line (trace) should appear on the screen. Adjust the intensity control to produce a visible but not too intense trace. Use the horizontal positioning control to move the left end of the line to the leftmost graduation of the graticule. Adjust the trace rotation control, if needed, for a proper horizontal trace.
6. Set the vertical input coupling switch, for channel 1, to the ground (GND) position and adjust the vertical position control to establish a zero volt reference position in the center of the screen.
7. Turn off the oscilloscope. Have someone change all the control and switch setting settings. Repeat steps 3 through 6.
8. Set your oscilloscope's control as follows:

Volts/Div.	1 V/div
Time/Div.	1 ms/div
Triggering	auto
Slope	+
Vert. Coupling	DC
9. Establish a zero volt reference position on the center of the oscilloscope's graticule.
10. With a variable dc power supply in the off position, connect your scope probe's ground lead to the negative output of the dc supply and the scopes positive lead to the positive output of the variable supply. Also, connect a DMM across the supply's output.
11. Adjust the dc output voltage to 1V as indicated on the DMM. Observe the change in the trace position on your scope. Adjust the output voltage to 2 and 3V respectively, noting the deflection

change on the scope.

12. Turn **off** the power supply. Disconnect the supply and turn **off** the oscilloscope.

CHARACTERISTICS OF A SINUSOID

Experiment # 2

Introduction

The experiment on the Characteristics of a Sinusoid is an educational endeavor designed to provide students and enthusiasts with a deep understanding of sinusoidal waveforms, which are fundamental in various fields of science and engineering, particularly in electronics, signal processing, and physics. A sinusoid, commonly known as a sine wave, is a mathematical curve that describes a smooth periodic oscillation. It is a pivotal concept in the study of wave phenomena and forms the basis for understanding more complex waveforms.

This experiment aims to explore the fundamental properties and parameters that define a sinusoidal wave, such as amplitude, frequency, phase, and wavelength. Participants will engage in practical activities that involve generating, observing, and analyzing sinusoidal waves, typically using function generators and oscilloscopes.

Through hands-on experience, participants will develop a practical understanding of how sinusoidal waves are used in various applications such as in AC power generation and transmission, signal processing, and communications. The experiment will also lay the groundwork for understanding more complex waveforms and their analysis, which is essential in many technological and scientific endeavors.

By the end of this experiment, students will have gained valuable insights into the behavior of sinusoidal waves and their importance in both theoretical and applied contexts, thereby enriching their knowledge base in fields related to electronics, physics, and engineering.

OBJECTIVES

1. Understanding the basic mathematical representation of a sinusoidal wave and how it correlates to the physical wave.
2. Learning how to generate a sinusoidal waveform using a function generator and how to manipulate its characteristics.
3. Observing and interpreting the characteristics of a sinusoidal wave on an oscilloscope.
4. Measuring the amplitude, frequency, and phase difference of sinusoidal waves and understanding their significance in real-world applications.
5. Exploring the concept of harmonics and how sinusoidal waves can be combined to form complex waveform

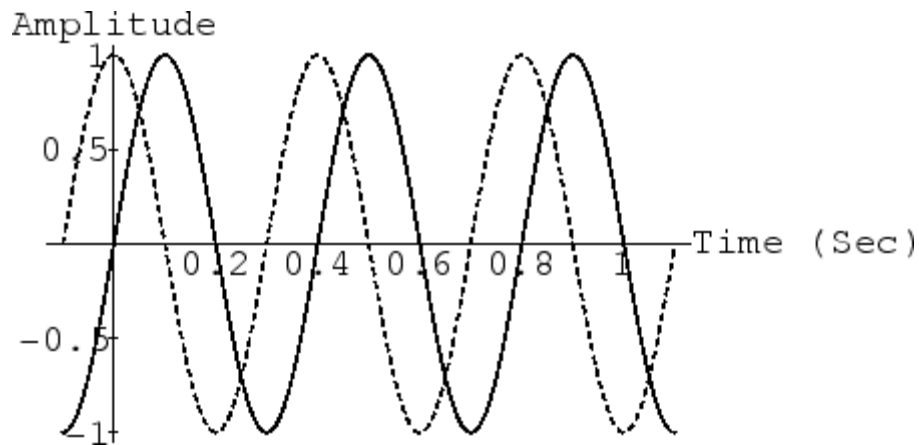


Figure 1.1

MATERIALS REQUIRED

- Personal Computer with Proteus

PROCEDURES

Given the AC sources

$$V_1 = 12/10^\circ \text{ V}$$

$$V_2 = 15/-100^\circ \text{ V}$$

$$V_3 = 10/10^\circ \text{ V}$$

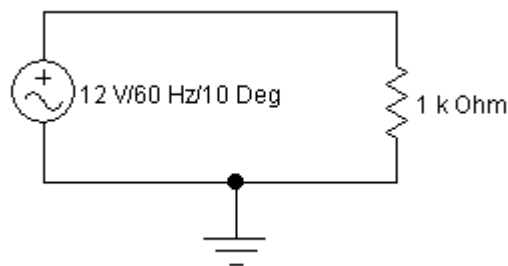


Figure 1.2 Circuit Diagram

- A. Amplitude, period, and frequency.
1. Create the circuit as in figure 1 for V_1 .
2. Connect channel A probe of the oscilloscope across the 1 kOhm load.
3. Set the volts/div and time base to fit one full wave on the graticule.
4. Click the switch for simulation. Observe how the sinusoid is generated.
5. Pause the simulation.
6. Estimate the amplitude of the signal with reference to the volts/division setting of the oscilloscope.

7. Move the red vertical line to a point where the signal moves upward and crosses the horizontal center line. Likewise, move the blue vertical line to another point where the signal moves upward and crosses the horizontal center line.
8. Modify the x-position of the signal so that the red line is collinear with one of the vertical lines on the grid.
9. Count the number of divisions separating the red and the blue line. Multiply this with the time base to get the period of the sinusoid.
10. Compute for the frequency of the sinusoid given the period.
11. On a separate sheet of paper, draw the face of the oscilloscope showing the sinusoid V_1 .

B. Phase relationships

1. Create a separate circuit as in figure 1 for V_1 , V_2 , and V_3 .
2. Connect channel A probe for V_1 and channel B probe for V_2 .
3. Click the switch for simulation. Observe how the sinusoid is generated.
4. Pause the simulation.
5. Observe which of the two signal leads.
6. Move the red vertical line to a point where V_1 moves upward and crosses the horizontal center line. Likewise, move the blue vertical line to a point where V_2 moves upward and crosses the horizontal center line.
7. The time difference between V_1 and V_2 is shown on the lower right corner of the graticule. Using this value compute for the phase difference between V_1 and V_2 .
8. On a separate sheet of paper, draw the face of the oscilloscope showing the sinusoid V_1 and V_2 .
9. Repeat steps 1 to 8 for V_1 and V_3 .

PHASE SHIFTING

Experiment # 3

Introduction

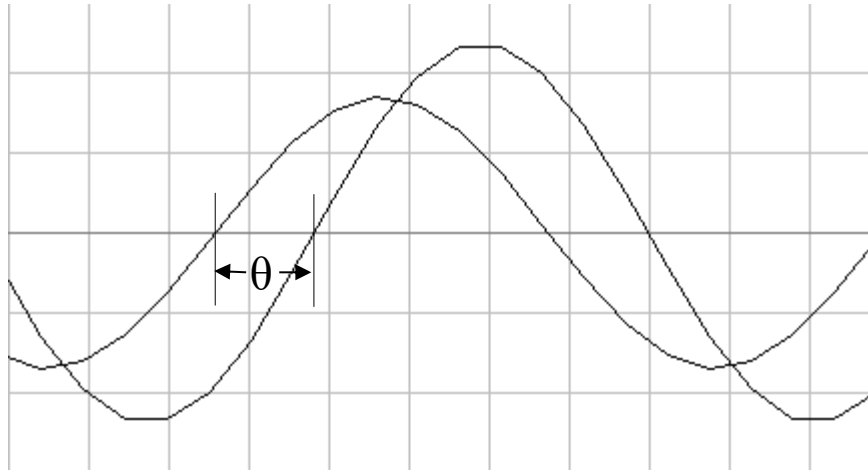
The experiment on Phase Shifting is a critical educational exploration aimed at understanding the concept of phase in waveforms, specifically focusing on the phenomenon of phase shifting. Phase shifting refers to the alteration of the phase of a waveform relative to another waveform or to its own initial state. This concept is fundamental in various fields such as signal processing, communications, electronics, and physics, where the relationship between different waveforms and their phase alignment is crucial.

This experiment introduces participants to the theoretical and practical aspects of phase shifting. It involves the use of electronic equipment such as function generators, oscilloscopes, and potentially phase shifters, to practically demonstrate how the phase of a signal can be altered and measured.

OBJECTIVES

1. Understanding the Concept of Phase: Participants will learn what phase is in the context of waveforms, and how it relates to the concepts of frequency and wavelength.
2. Theoretical Foundations of Phase Shifting: This includes learning the mathematical principles that underpin phase shifting, including how phase differences are represented and calculated.
3. Practical Demonstration of Phase Shift: Using electronic equipment, participants will observe how phase shifting occurs in real-time. This will involve generating waveforms and intentionally altering their phase.
4. Measuring Phase Shift: Learning how to use an oscilloscope to measure the phase difference between two waveforms, which is a critical skill in many electronic and signal processing applications.
5. Exploring Applications of Phase Shifting: Understanding how phase shifting is used in real-world scenarios, such as in communication systems (e.g., phase modulation), signal processing, and in the study of wave interference.
6. Analyzing the Impact of Phase Shift: Participants will learn how a phase shift affects the properties of a system or a signal, such as in the creation of constructive or destructive interference patterns.
7. Critical Thinking and Problem Solving: Encouraging participants to think critically about how phase shift can be used to solve practical problems in engineering and physics.

A phase-shifting circuit is often employed to correct an undesirable phase shift already present in a circuit or to produce special desired effects. An RC circuit is suitable for this purpose because its capacitor causes the circuit current to lead the applied voltage (Alexander and Sadiku, 2007).



The phase-shift between two sinusoids can be calculated using the equation

$$\theta = 2\pi ft$$

Where, f is the operating frequency and t is the time difference between two sinusoids. This shows that the amount of phase shift depends on the value of R , C , and the operating frequency.

MATERIALS REQUIRED

- Personal Computer with Proteus

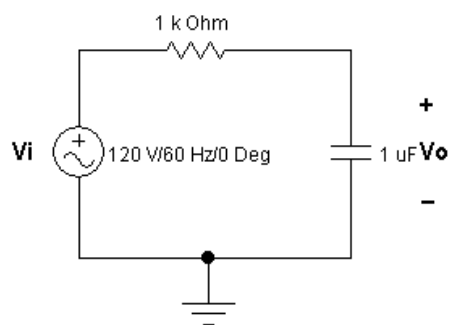


Figure 1

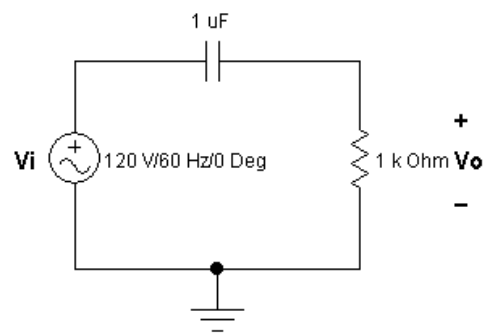


Figure 2

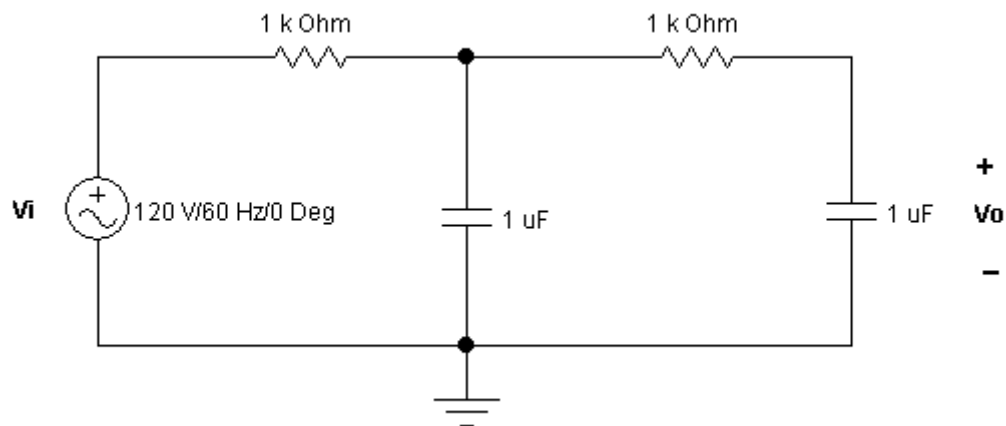


Figure 3

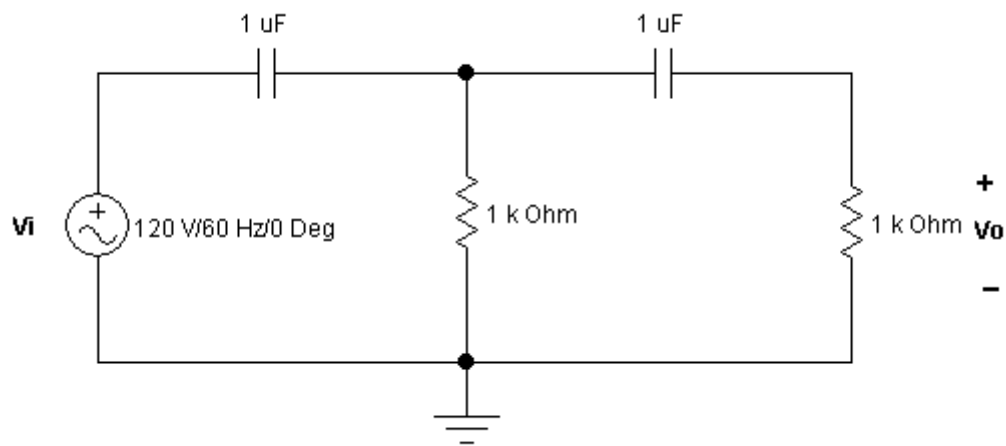


Figure 4

PROCEDURES

1. Using Proteus assemble the circuits shown in the figure above.
2. View the graph of the sinusoids using the oscilloscope.
3. Adjust the necessary controls to fit the sinusoid on the oscilloscope screen.
4. Measure the phase shift between V_i and V_o in figures 1 and 2.
5. Using the values of the components in figures 1 and 2, calculate the phase shift between V_i and V_o .
6. Draw the graph of the sinusoids.
7. Observe the amplitudes of the sinusoids.
8. Double the frequency of the source.
9. Repeat steps 4 – 7.
10. Assemble figures 3 and 4.
11. Repeat steps 2 – 7.

Average and RMS Values

Experiment # 4

Introduction

The experiment on Average and RMS (Root Mean Square) Values is a fundamental educational pursuit designed to deepen the understanding of these crucial statistical measures in the context of electrical engineering and physics. This experiment focuses particularly on their application to analyzing electrical signals, which is paramount in fields such as power engineering, signal processing, and electronics.

Average and RMS values are critical in describing the characteristics of varying electrical signals, especially when dealing with alternating current (AC) circuits. The average value (often referred to as the mean) of a waveform gives a basic statistical measure of its central tendency over a period. In contrast, the RMS value provides a measure of the effective magnitude of a varying signal, akin to the magnitude of a direct current (DC) that would deliver the same power to a load as the AC signal.

Learning Objectives:

1. **Understanding Average and RMS Concepts:** Participants will learn the theoretical basis of average and RMS values, including their mathematical definitions and physical significance.
2. **Measurement Techniques:** This includes learning how to calculate and measure the average and RMS values of various waveforms, particularly sinusoidal AC signals.
3. **Practical Application Using Electronic Instruments:** Participants will use tools such as multimeters, oscilloscopes, and waveform generators to practically observe and measure these values in real electrical signals.
4. **Analyzing Different Waveforms:** The experiment will not be limited to sinusoidal waves; participants will also explore how average and RMS values apply to other forms of waveforms, such as triangular and square waves.
5. **Comparing Average and RMS Values:** Understanding the differences between these two measurements and their respective implications in analyzing electrical circuits and systems.
6. **Real-World Applications:** Participants will explore how average and RMS values are used in practical scenarios, such as in power delivery, signal processing, and in the design and analysis of electrical circuits.
7. **Critical Thinking and Interpretation:** Encouraging participants to interpret the results and understand the significance of these values in practical electrical and electronic applications.

By the end of this experiment, participants should have a comprehensive understanding of average and RMS values, their calculations, and their critical roles in electrical and electronic engineering. This knowledge is indispensable for students and professionals involved in the analysis and design of electrical systems and signals.

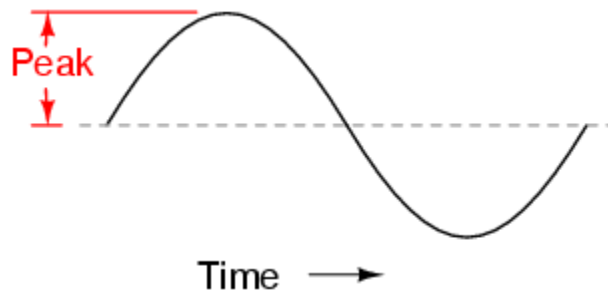
Waveforms of voltage and current that vary periodically with time may be characterized by their average value or their root mean square (rms) value. The latter is used to determine the power supplied,

dissipated, or stored by a circuit element. Some of the measuring instruments you will use respond to average values of voltage or current, while others respond to rms values.

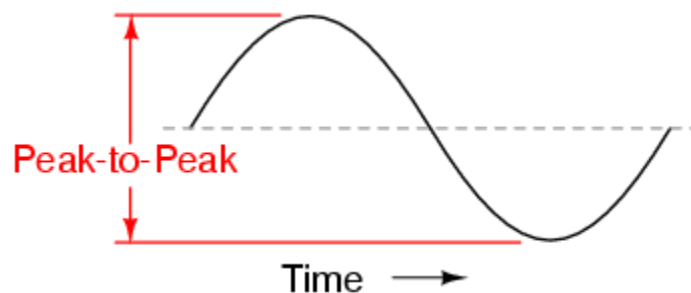
BACKGROUND

Measurements of AC signal:

- *Amplitude* is a nonnegative scalar measure of a wave's maximum magnitude of oscillation. In electrical engineering it may be thought of as the maximum absolute value reached by a voltage or current waveform as measured from the center of the oscillation. An amplitude **measurement** may be reported as peak, peak-to-peak, average, or RMS.
- *Peak amplitude* is the height of an AC waveform as measured from the center of the oscillation to the highest positive or lowest negative point on a graph. Also known as the *crest* amplitude of a wave.



- *Peak-to-peak amplitude* is the total height of an AC waveform as measured from maximum positive to minimum negative peaks (the highest peak to the lowest valley) on a graph of the waveform. Often abbreviated as "P-P", e.g., V_{p-p} or V_{pp} .



- *Average value* is the arithmetic "mean" of a waveform's values over one cycle. The average value of any waveform with equal-area portions above and below the "zero" line on a graph is zero. However, often as a practical measure of amplitude, a waveform may be characterized by its *average absolute value*, calculated as the arithmetic mean of the *absolute values* of the waveform.
- "RMS" stands for *Root Mean Square*, and is a way of expressing an AC quantity of voltage or current in terms functionally equivalent to DC. For example, 10 volts AC RMS is the amount of AC voltage that would produce the same amount of heat dissipation across a resistor of given value as a 10 volt DC power supply. Also known as the "equivalent" or "DC equivalent" value of an AC voltage or current.
- Analog, electromechanical meter movements respond proportionally to the *average absolute value* of an AC voltage or current. When RMS indication is desired, the meter's calibration must be adjusted accordingly, usually by applying a constant multiplicative factor assumed for a sinusoidal wave. This means that the accuracy of an electromechanical meter's RMS indication is dependent on the purity of the waveform and whether it is the same wave shape as the waveform used in calibrating.

In this experiment, we will work with periodic waveforms having period T and amplitude V_m . Specifically, we will work with the following three types of waveforms:

1. A sinusoidal wave voltage:

$$v(t) = V_m \cos(2\pi t/T)$$

2. A square-wave voltage:

$$v(t) = V_m \text{ for } 0 \leq t \leq \frac{T}{2}; T \leq t \leq \frac{3T}{2}; \text{etc.}$$

3. A triangular-wave voltage:

$$v(t) = 4V_m[(t - nT)/T] \text{ for } (n - \frac{1}{4}T) \leq t \leq (n + \frac{1}{4}T)$$

$$v(t) = V_m \left[\left(t - \left(n + \frac{1}{4} \right) nT \right) / T \right] \text{ for } (n + \frac{1}{4}T) \leq t \leq (n + \frac{3}{4}T)$$

AC voltage and current waveforms are further depicted in a number of ways. Two of the most common and useful are the average and the root-mean-square (rms) values.

The average value of a time-varying waveform $x(t)$ with period T is given by:

$$X_{ave} = \frac{1}{T} \int_0^T x(t) dt$$

The root-mean-square value, useful for power calculations, is defined by:

$$X_{rms} = \sqrt{\frac{1}{T} \int_0^T x^2(t) dt}$$

MATERIALS REQUIRED

Power Supply:

- Power source for oscilloscope
- 0-15 V dc variable supply

Instruments:

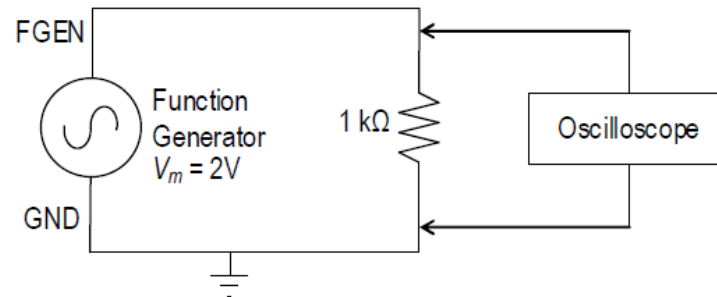
- Oscilloscope
- DMM

Miscellaneous:

- Assorted leads for connecting to the oscilloscope terminals and jacks
- Resistor Decade Box

PROCEDURE

1. Connect the output terminals of the Function Generator to a Resistance Decade Box



2. Set the decade box to $1\text{ k}\Omega$ resistance as the load on the output of the Function Generator.
3. Turn on the Function Generator and use the oscilloscope to observe the voltage drop across the $1\text{ k}\Omega$ resistance. *Remember that the oscilloscope's ground is the same as the Function Generator's ground.*
4. Select the sine wave output option of the Function Generator
 - a. Set the frequency of the function generator to 1 kHz
 - b. Observe the waveform on the oscilloscope using the CH0 BNC connector.
 - c. Connect the SYNC signal (located directly below the FGEN signal row) to the Trigger BNC input on the left side on the workstation. Change the Trigger type to "Digital".
 - d. Adjust the DC offset control of the Function Generator to obtain a zero DC level in the output.
 - e. Use the Function Generator's output level control (Amplitude control knob/numerical input) to obtain a 2.0 Volt peak value (4.0 Vpp) on the scope measurement.
 - g. Use the digital multimeter (DMM) AC Voltage [V~] function to measure the Function Generator [FGEN] voltage by connecting the V Ω and COM banana jacks to FGEN and GND respectively. Make sure to include DC offset values in the RMS calculation

CAPACITORS AND SERIES RC CIRCUITS

Experiment # 5

Introduction

The experiment on Capacitors and Series RC Circuits is an essential educational endeavor in the field of electronics and electrical engineering. This experiment provides a practical understanding of capacitors, one of the fundamental passive components in electronic circuits, and their behavior in series with resistors (forming RC circuits). Capacitors are widely used for various functions such as energy storage, filtering, and timing applications. Understanding their characteristics in combination with resistors is crucial for anyone involved in circuit design and analysis.

In this experiment, participants will construct and analyze series RC circuits, which consist of a resistor (R) and a capacitor (C) connected in series. The focus will be on understanding how these components interact and how their combination affects the circuit's overall behavior, particularly in response to varying input signals.

Learning Objectives:

1. Fundamentals of Capacitors and Resistors: Participants will gain a foundational understanding of how capacitors and resistors function individually, including their basic properties and characteristics.
2. Constructing Series RC Circuits: Participants will learn how to assemble a series RC circuit and the importance of component orientation and connection.
3. Time Constant and Transient Response: One of the key learning points will be understanding the time constant in RC circuits, which determines how quickly the circuit responds to changes in voltage. Participants will explore the concept of transient response, observing how the capacitor charges and discharges.
4. Voltage and Current Behavior: Through practical exercises, participants will observe and analyze how voltage across and current through the capacitor and resistor change over time in response to an applied voltage.
5. Frequency Response of RC Circuits: Participants will explore how the RC circuit behaves at different frequencies, understanding the concepts of filtering and phase shift.
6. Use of Electronic Measuring Instruments: Hands-on experience with instruments like multimeters and oscilloscopes to measure and visualize circuit parameters.
7. Practical Applications: Discussing real-world applications of RC circuits, such as in timing circuits, filters, and signal processing.
8. Analytical Skills Development: Encouraging analytical thinking by having participants predict, observe, and explain the behavior of the circuit under various conditions.

By the end of this experiment, participants should be able to understand the basic operation of capacitors and series RC circuits, how to build and analyze such circuits, and appreciate their significance in various

electronic applications. This experiment is vital for anyone looking to deepen their knowledge in electronics, particularly in areas involving circuit design and signal processing.

PRE-LAB:

Reading:

1. Read and study the Background section of this Laboratory.
2. Research from the internet and write a summary of the following:
 - a. Avoiding Grounding Errors with Oscilloscope
 - b. Voltage Measurement for a two channel oscilloscope
 - c. Phase-Angle Measurement for a two channel oscilloscope

Written:

1. In your lab notebook sketch the circuit diagram to be used in the procedure and prepare Tables 3.1 and 3.2 to record data.
2. Sketch the impedance and voltage phasors diagrams (as in Figure 3.2) you would get at 500 Hz for the circuit in Figure 3.3.
3. Make a table of the magnitudes and phase angles. You will use this to check that your experimental setup is correct.

MATERIALS REQUIRED

Power Supply:

- Power source for oscilloscope

Instruments:

- Oscilloscope
- DMM
- Function Generator

Miscellaneous:

- Assorted leads for connecting to the oscilloscope terminals and jacks
- Resistor Decade Box
- Capacitors, 0.01 μf , qty 2 (discrete capacitors)

BACKGROUND

A capacitor is formed whenever two conductors are separated by an insulating material. Consider the simple example of two parallel conducting plates separated by a small gap that is filled with an insulating material (vacuum, air, glass, or other dielectric). If a potential difference exists between the two plates, then an electric field exists between them, and opposite electric charges will be attracted to the two plates. The ability to store that electric charge is a fundamental property of capacitors. The larger the plates, the more charge can be stored. The closer the plates, the more charge can be stored...at least until the charges leap the gap and the dielectric breaks down. If a voltage source is connected across a capacitor, charge will flow in the external circuit until the voltage across the capacitor is equal to the applied voltage. The charge that flows is proportional to the size of the capacitor (its "capacitance") and to the applied voltage. The relationship is given by the equation

$$Q = CV$$

where Q is the charge in coulombs, C is the capacitance in farads, and V is the applied voltage in volts.

Capacitors in Series

Electric current I is the amount of charge that flows per unit time; that is, $I = Q/T$. Thus, the total charge that flows through a circuit (or a capacitor) is $Q = IT$. So, if two capacitors are connected in series and a voltage is applied across the pair, the same current, and therefore the same charge, must flow through both capacitors, and the total voltage V_T must be divided across both capacitors:

$$V_T = V_1 + V_2 = \frac{Q}{C_1} + \frac{Q}{C_2} = Q \left(\frac{1}{C_1} + \frac{1}{C_2} \right) = \frac{Q}{C_T}$$

where V_1 and V_2 are the voltages across the capacitors with capacitances C_1 and C_2 . Thus, C_T , the total capacitance of two capacitors in series, is found by

$$\frac{1}{C_T} = \frac{1}{C_1} + \frac{1}{C_2}$$

Capacitors in Parallel

Connecting capacitors in parallel is effectively the same as making a single capacitor's plates larger, and therefore able to hold more charge for a given applied voltage. This simple view is borne out if one analyzes the flow of charge through a parallel array of capacitors connected to a voltage source. The result of such analysis is that capacitances in parallel add directly:

$$C_T = C_1 + C_2 + C_3 + \dots$$

Time Constant

If a voltage V_0 is applied to a capacitor C connected in series with a resistor R , the voltage across the capacitor gradually increases. The rate at which the capacitor's voltage changes is characterized by a "time constant", τ :

$$\tau = RC$$

- τ is the time required for the voltage on the capacitor to rise from 0 to $0.632 V_0$.
- τ is also the time required for the voltage of a fully charged capacitor to fall from V_0 to $0.368 V_0$.
- The number $0.368 = e^{-1}$ and the number $0.632 = (1 - e^{-1})$.

Capacitive Reactance

Reactance is a characteristic exhibited by capacitors and inductors in circuits with time-varying voltages and currents, such as common sinusoidal AC circuits. Like resistance, reactance opposes the flow of electric current and is measured in ohms. Capacitive reactance X_C can be found by the equation:

$$X_C = \frac{1}{2\pi fC}$$

where f is the frequency of the applied voltage or current and C is the capacitance in farads. As with resistance, reactance obeys Ohm's law:

$$V_C = I_C X_C \text{ or } X_C = \frac{V_C}{I_C}$$

If a sinusoidal voltage is applied across a *resistor*, the current through the resistor is *in phase* with the voltage. That is not true for a capacitor. If we connect a *capacitor* across a sinusoidal voltage, the maximum current flows through the capacitor when the voltage's rate of change is maximum (i.e., at $V=0$), and diminishes as the voltage on the capacitor increases, until finally the current is zero when the voltage is at maximum and its derivative is zero. At that instant, the maximum possible charge for the applied voltage is stored in the capacitor, and so the flow of charge (i.e., the current) stops. The current and the voltage have exactly the same frequency, but the current through the capacitor is *leading* the voltage by $\frac{1}{4}$ cycle - 90° or $\pi/2$ radians. Figure 3.1 illustrates this relationship.

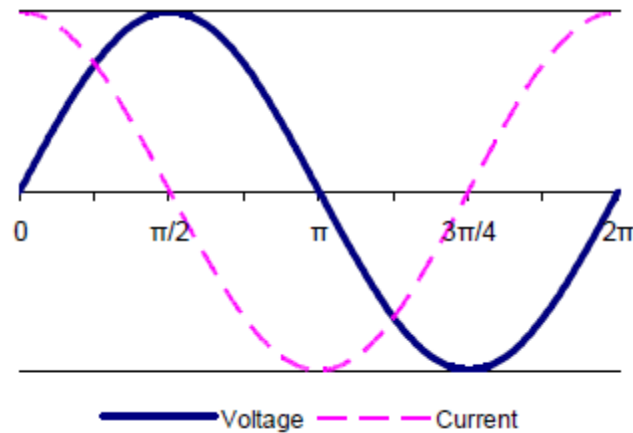


Fig 3.1

Phasors

When a sinusoidal voltage at frequency f drives a circuit that contains only linear elements, the waveforms throughout the circuit are also sinusoidal and at the same frequency. To understand the relationships among the sinusoidal voltages, currents, and impedances, we represent the various waveforms as two-dimensional vectors called *phasors*. A phasor is a complex number used to represent a sinusoidal wave, taking into account both its amplitude and phase angle. As a complex number, a phasor has “real” and “imaginary” components, but like any two-dimensional vector, it can be drawn simply on ordinary XY axes, with the “real” axis in the usual X direction and the “imaginary” axis in the usual Y direction. Such phasor drawings are very helpful in analyzing circuits and understanding the relationships of the various voltages and currents. The algebra of complex numbers can then be used to perform arithmetic operations on the sinusoidal waves. Make no mistake: adding voltages or currents in an AC circuit without taking account of phase angles will lead to confusing and wrong results. A series RC circuit is illustrated in Figure 3.2(a) and a phasor diagram of its impedances is shown in Figure 3.2(b). The vector for resistance R is shown along the Real (X) axis, while the reactance XC is shown in the negative Imaginary (Y) axis, since its voltage trails its current by 90° . The vector sum of R and XC is labeled Z and has magnitude $5\text{ k}\Omega$. Therefore, the magnitude of the current through the circuit is $V_s / Z = 1.0\text{ mA}$. Notice that the current phasor is in the same as the direction of the voltage across the resistor, because voltage and current are in phase for a resistor. The current is the same throughout the series circuit, including the capacitor, and the voltage across the capacitor trails the current by 90° . The source voltage is the **vector sum** of the voltages across the resistor and the capacitor, as is illustrated in the phasor diagram, Figure 3.2(c).

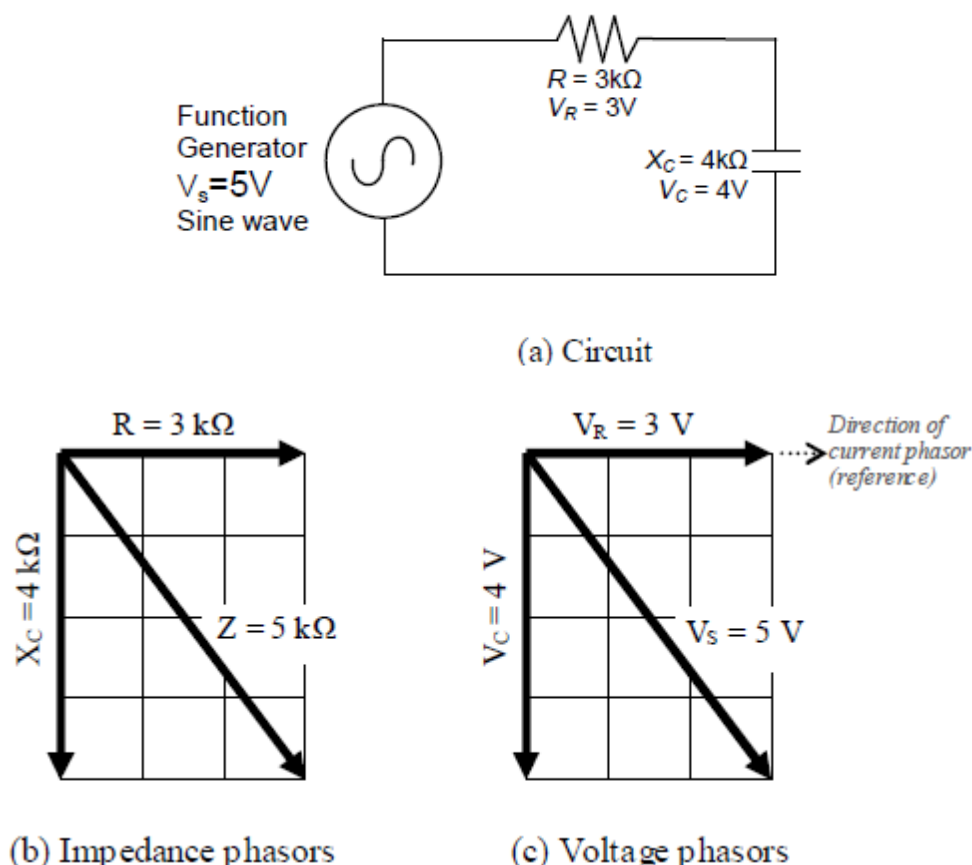


Fig 3.2

PROCEDURE

- Set $10 k\Omega$ on the resistor decade box and measure its resistance using the digital multimeter (DMM). Record the value in Table 3.1.
- Select two $0.01 \mu F$ capacitors. Identify them as C_1 and C_2 .
 - Measure the capacitances using the DMM by connecting the capacitor leads to the DUT+ and DUT- pin sockets. (DUT means "device under test".) Record the values in Table 3.1
 - Connect the capacitors in **parallel** and measure the combined capacitance using the DMM and record the value in Table 3.1. Wait for the measurement to stabilize before recording. Calculate the theoretical prediction for the parallel capacitance and record it in Table 3.1. Compare the theoretical and measured values. Does the measurement tend to confirm the theoretical prediction? Explain.
 - Connect the capacitors in **series** and measure the combined capacitance using the DMM and record the value in Table 3.1. Wait for the measurement to stabilize before recording. Calculate the theoretical prediction for the series capacitance and record it in Table 3.1. Compare the theoretical and measured values. Does the measurement tend to confirm the theoretical prediction? Explain.
 - Select capacitor C_1 . Compute its theoretical reactance X_{C1} at a frequency $f = 500$ Hz. Record the value in Table 3.1

Table 3.1

	Listed or Theoretical Value	Measured Value
R		
C₁		
C₂		
C₁_C₂ parallel		
C₁_C₂ series		
X_{C1} (500 Hz)		

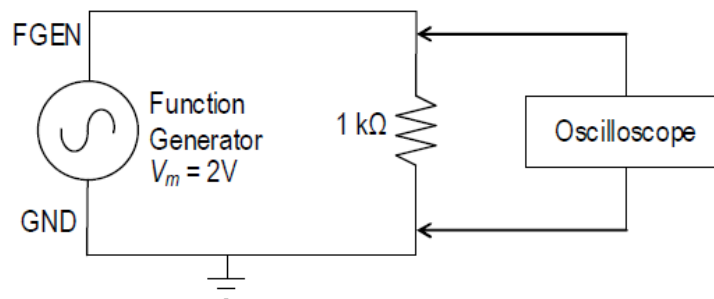
For steps 3 through 6, record the results of the measurements in Table 3.2. Throughout the experiment measure all voltages as peak-to-peak values.

Table 3.2

Frequency, Nominal	Frequency, Actual	V _s , Actual	V _{C1}	V _R	I _{PP}	Φ	X _{C1}	C ₁	Z _T
500 Hz									
1000 Hz									
2000 Hz									
4000 Hz									
8000 Hz									

PROCEDURE

- Connect the output terminals of the Function Generator to a Resistance Decade Box



- Set the decade box to 1 kΩ resistance as the load on the output of the Function Generator.
- Turn on the Function Generator and use the oscilloscope to observe the voltage drop across the 1 kΩ resistance. *Remember that the oscilloscope's ground is the same as the Function Generator's ground.*
- Select the sine wave output option of the Function Generator
 - Set the frequency of the function generator to 1 kHz
 - Observe the waveform on the oscilloscope using the CH0 BNC connector.
 - Connect the SYNC signal (located directly below the FGEN signal row) to the Trigger BNC input on the left side on the workstation. Change the Trigger type to "Digital".
 - Adjust the DC offset control of the Function Generator to obtain a zero DC level in the output.

- e. Use the Function Generator's output level control (Amplitude control
 - f. knob/numerical input) to obtain a 2.0 Volt peak value (4.0 Vpp) on the scope measurement.
 - g. Use the digital multimeter (DMM) AC Voltage [V~] function to measure the Function Generator [FGEN] voltage by connecting the V Ω and COM banana jacks to FGEN and GND respectively. Make sure to include DC offset values in the RMS calculation
9. Repeat the steps from 4a to 4f using a triangular wave output from the Function Generator.
 10. Repeat the steps from 4a to 4f using a square wave (50% duty factor) output from the Function Generator.
 11. Adjust the DC offset control to add a 2 Volt DC offset to the original 1 kHz sine wave.
 12. Now the waveform should vary from 0 to 4 Volts. Sketch this waveform in your lab notebook. Use the DMM to measure the AC RMS voltage and the DC voltage across the resistance. Record the values. Repeat this same step for triangular wave and square wave.
 13. Compute the capacitor's reactance X_{C1} from $X_{C1} = V_{C1}/I_{PP}$. Compute C_1 from the measured X_{C1} and compare to your earlier measurement.
 14. Compute the total impedance Z_{Total} by applying Ohm's law to the circuit. Use the supply voltage set in step 3 and the current found in step 4. Remember, the impedance has both a magnitude and a phase angle (measured relative to the resistor).
 15. Repeat steps 3 to 10 (resetting V_S if necessary) for the following frequencies: 1000, 2000, 4000, 8000 Hz.
 16. Draw impedance and voltage phasors (as in Figure 3.2) for frequency $f = 1000$ Hz.
 17. Draw impedance and voltage phasors for frequency $f = 4000$ Hz.
 18. The phasor diagrams at various frequencies show how the impedances, and therefore the voltages, change with frequency. To better see the net effect on the circuit, graph V_{C1} and V_R versus frequency for the values in Table 3.2. Label the curves.

INDUCTORS AND SERIES RL CIRCUITS

Experiment # 6

Introduction

The experiment on Inductors and Series RL Circuits offers a comprehensive exploration into the world of inductors, one of the fundamental components in electronics, and their behavior in series with resistors, forming RL circuits. Inductors, known for their ability to store energy in a magnetic field, are pivotal in various applications ranging from power supply design to radio frequency tuning. Understanding the dynamics of how inductors work in combination with resistors is critical for students and professionals engaged in electronic circuit design and analysis.

This experiment involves the construction, observation, and analysis of series RL circuits, which consist of a resistor (R) and an inductor (L) connected in series. The primary focus is on understanding the interaction between these components and how their combination impacts the overall behavior of the circuit, especially in response to different input signals.

Learning Objectives:

1. **Understanding Inductors and Resistors:** Participants will delve into the fundamental properties and functions of inductors and resistors, learning how each component behaves individually within a circuit.
2. **Building Series RL Circuits:** This objective involves guiding participants through the process of assembling a series RL circuit and emphasizing the importance of component orientation and connections.
3. **Exploring Time Constant and Transient Response:** A key learning point will be the concept of the time constant in RL circuits, which determines the circuit's response speed to voltage changes. The experiment will cover the transient response of the circuit, observing the inductor's charging and discharging behavior.
4. **Analyzing Voltage and Current Behavior:** Participants will conduct experiments to observe and interpret how voltage across and current through the inductor and resistor change over time in response to an applied voltage.
5. **Frequency Response of RL Circuits:** The experiment will explore how RL circuits respond to different signal frequencies, focusing on concepts such as inductive reactance and phase shift.
6. **Utilization of Measurement Instruments:** Hands-on experience with electronic instruments, such as multimeters and oscilloscopes, will be integral for measuring and visualizing circuit parameters.
7. **Real-World Applications:** Discussion of practical applications of RL circuits in areas like filtering, signal processing, and electromagnetic field generation.
8. **Developing Analytical Skills:** Encouraging participants to predict, observe, and rationalize the circuit behavior under various conditions to enhance their analytical and critical thinking abilities.

Upon completion of this experiment, participants will have a thorough understanding of the behavior of inductors in series with resistors, the construction and analysis of RL circuits, and their relevance in various electronic applications. This knowledge is essential for those involved in electronics, particularly in fields related to circuit design, electromagnetism, and signal processing.

PRE-LAB:

Reading:

1. Study the Background section of this Laboratory experiment.
2. Review previous assignments on *Avoiding Grounding Errors with Oscilloscope*, and *Phase-Angle Measurement* for a two-channel oscilloscope.

Written:

1. In your lab notebook sketch the circuit diagram to be used in the procedure.
2. Prepare tables to record the data.
3. Sketch the impedance and voltage phasors diagrams (as in Figure 4.2) you would get at 10 kHz for the circuit in Figure 4.3, using nominal values for the components. Make a table of the magnitudes and phase angles. You will use this to check that your experimental setup is correct.

MATERIALS REQUIRED

Power Supply:

- Power source for oscilloscope
- 0-15 V dc variable supply

Instruments:

- Oscilloscope
- DMM
- Function Generator

Miscellaneous:

- Assorted leads for connecting to the oscilloscope terminals and jacks
- 5 k Ω Resistor Decade Box
- 5 mH Inductor
 - Components may be discrete or via decade substitution boxes

BACKGROUND

When a current passes through a wire, a magnetic field is generated around the wire. The magnetic field results from the movement of electric charge and is proportional to the magnitude of the current. Turning the wire into a coil concentrates that magnetic field, with the field of one turn of the coil reinforcing another. Such a coil is called an *inductor*. From an electrical point of view the especially interesting fact is that if the current in the wire changes, the magnetic field will react to try to keep the current constant. This property of inductors is described by Lenz's law.

An inductor's response to changes of current is called *inductance*. Inductance opposes changes in *current*, just as capacitance opposes changes in *voltage*. Inductance is the electric current equivalent of inertia in mechanical systems.

Inductance is measured in *henries*. One henry is the amount of inductance present when one volt is generated as a result of a current changing at the rate of one ampere per second.

When inductors are connected in series, the total inductance is the sum of the individual inductors. This is similar to the way resistors in series add. When inductors are connected in parallel, the inductances add in the same way that parallel resistors add. However, an additional effect can appear in inductance circuits that is not present with resistors. This effect is called *mutual inductance* and is caused by the interaction of the magnetic fields of neighboring inductors. Mutual inductance can either increase or decrease the total inductance, depending on the orientation of the interacting inductors.

Inductance circuits have a time constant associated with them, just as do capacitance circuits, except that the curve of interest for inductors is the current, rather than the voltage. The time constant τ for inductors is

$$\tau = \frac{L}{R}$$

where τ is in seconds; L is in henries; R is in ohms.

The voltage induced across the inductor is a maximum when the *change* in current is a maximum. When a sinusoidal current is applied to an inductor, the largest induced voltage appears across the inductor when the current is passing through zero. At the peaks of the applied current, the slope is zero and the current is not changing, so the induced voltage is zero. Therefore, the voltage that appears across an inductor leads the current in the inductor by 1/4 cycle.

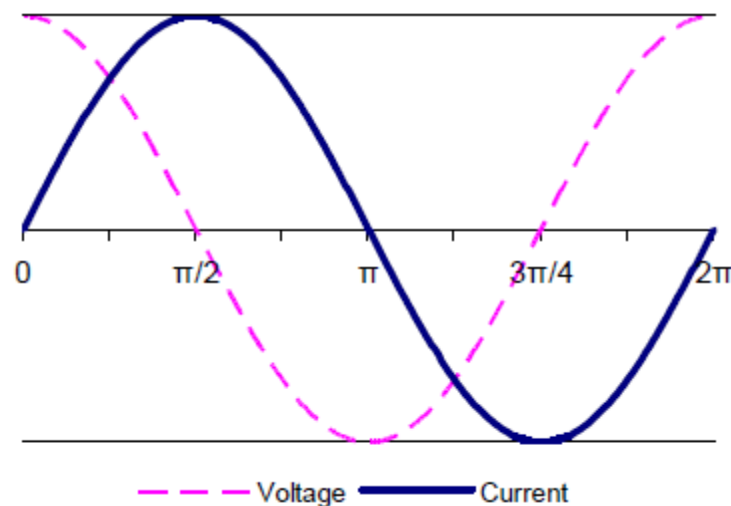


Figure 4.1

Inductive reactance

As the frequency of the sine wave increases, the rate of change of the current also increases, and so the induced (reacting) voltage across the inductor increases. As a result, the net current through the inductor decreases. That means, the inductor's *reactance* increases with frequency. The inductive reactance is given by

$$X_L = 2\pi fL$$

As with capacitors and resistors, Ohm's law can be applied to inductive circuits:

$$X_L = \frac{V_L}{I_L}$$

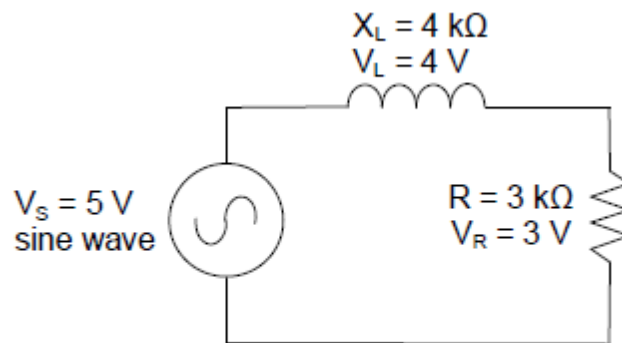
Series RL circuits

When a sine wave is applied to a series circuit of linear components (resistors, capacitors, and inductors), the phase relationships between current and voltage depend on the types of components. The current and voltage are always in phase across an ideal resistor. The current through a capacitor leads the voltage across the capacitor by 90° . The voltage across an ideal inductor leads the current through the inductor by 90° .

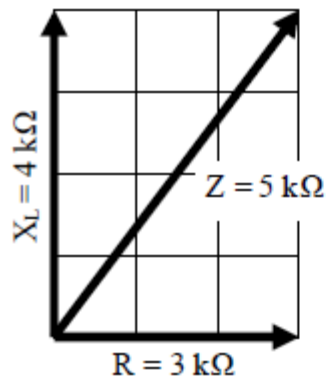
A common memory aid for these relationships is "*ELI the ICE man*", where *E* represents voltage (E is short for "electromotive force", which is another term for voltage), *I* represents current, *L* represents inductance, and *C* represents capacitance.

From Kirchhoff's current law (KCL), we know that the current is the same throughout a series circuit. Since the current and voltage are in phase for a resistor, we can determine the phase of the current by measuring the phase of the voltage across the resistor. This is commonly done by using an oscilloscope to compare the voltage from the source to the voltage across the resistor.

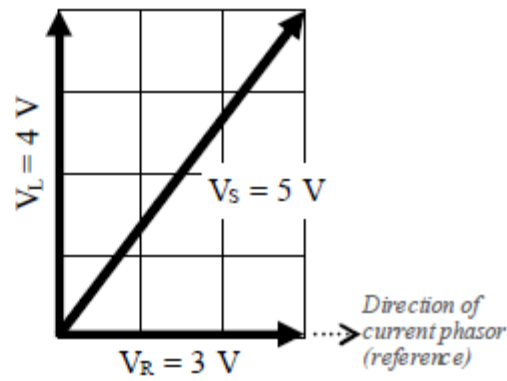
Consider the series RL circuit shown in Figure 4.2(a). Graphical representations for the impedance phasors are shown in Figure 4.2(b). As with the series RC circuit, the total impedance of the series RL circuit is obtained by the vector sum of the impedance phasors. (Because the vectors are in the complex plane, this summation can also be done equivalently by summing the complex numbers.)



(a) Circuit



(b) Impedance phasors



(c) Voltage phasors

Figure 4.2

In this example, 5V are applied and the total impedance is 5 k Ω , so the total current is 1.0 mA. Because the current is the same in all components of the series circuit, we use its direction through the resistor for the phase reference direction. Multiplying the impedance phasors by the current gives the voltage phasors, as shown in Figure 4.2(c).

In this experiment you will use an oscilloscope to measure the phase angles in the circuit. Because of the resistance of the wires in an inductor, actual inductors may have enough resistance to affect the phase angle. To minimize that effect in this experiment, we will use a series resistor that is large compared to the inductor's resistance.

PROCEDURE

Use peak-to-peak readings for all voltage and current measurements in this experiment.

1. Construct a table for recording experimental data:

R	L	F	V_S (gen)	V_S (Osc)	V_R

V_L	I	X_L	Z_T	ϕ_{meas}	ϕ_{calc}

2. Select a 5 k Ω resistor or set that value on the resistor decade box. Measure its resistance using the digital multimeter (DMM) and record the measured value.
3. Select a 50 mH inductor or set that value on the inductor decade box. Measure its inductance L and winding resistance R_W using the DMM and record the value. To measure inductance requires connecting the leads to DUT+ and DUT–, as you did for the capacitor.
4. Construct the circuit shown in Figure 4.3. With the circuit connected, adjust the function generator's frequency f to 10 kHz with an output voltage of 4.0 VPP. Record the actual values of f and V_S (generator).

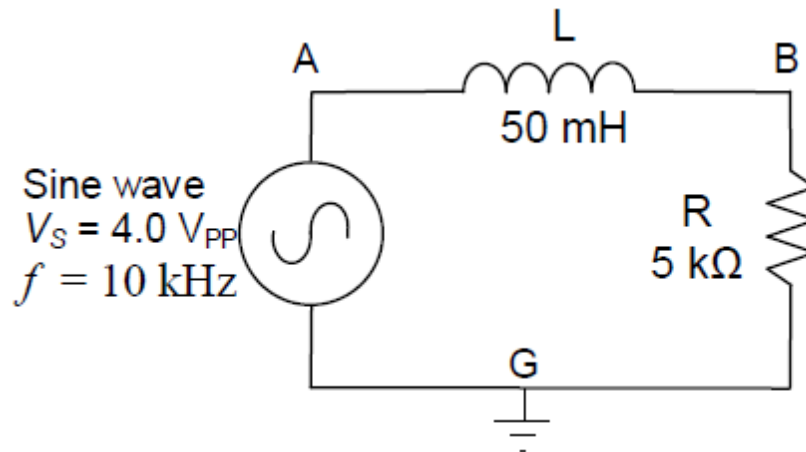


Figure 4.3 Series RL Circuit

5. Using the 2-channel oscilloscope, connect CHANNEL 0 to measure V_S , the voltage across the source (nodes A and G). Connect CHANNEL 1 to measure V_R , the voltage across the resistor (nodes B and G). Be sure that the ground leads are both connected to the same point (node G). Record V_S (Oscilloscope) and V_R .
6. With the oscilloscope still connected as in step 5, measure the phase angle ϕ between V_R and V_S . The method is described in Appendix C, section "Phase Angle Measurement", subsection "For an oscilloscope with two vertical inputs". Set the TRIGGER to EDGE and trigger on CHANNEL 0. You may use the AUTOSCALE button to ensure the waveforms are approximately the same height and press STOP to improve accuracy when taking cursor measurements. Record this value as ϕ_{meas} in the table.
7. **Interchange** R and L . [This is required because the ground connection of the NI-ELVIS oscilloscope input must be connected to circuit ground, and therefore, one of the leads of the device to be tested must also be connected to ground.] Connect CHANNEL 1 of the oscilloscope across the inductor L and measure V_L . Record the value.
8. Calculate the peak-to-peak current in the circuit by applying Ohm's law to the resistor.
That is, $= \frac{V_R}{R}$, Record the calculated current I in the table.
9. Calculate the inductive reactance X_L by applying Ohm's law to the inductor.

That is, $X_L = \frac{V_L}{I}$,

Record the calculated reactance in the table.

10. Calculate the total impedance Z_T by applying Ohm's law to the entire circuit.

$$Z_T = \frac{V_S}{I}$$

Record the value of Z_T in the table.

11. Using the values listed in the table, draw a diagram of the impedance phasors and a second diagram of the voltage phasors, as illustrated in Figure 4.2 of the Background section of this laboratory experiment.
12. Compute the phase angle ϕ between V_S and V_R . Recall that

$$\varphi = \arctan \left(\frac{V_L}{V_R} \right)$$

Record this value as φ_{calc} in the table, and compare it to the value measured on the oscilloscope.

13. Using the data in the table, convert the current I and source voltage V_S to RMS values. Then draw plots of the power phasor diagrams at the frequency of 10k Hz. Determine the real power, the reactive power, and the apparent power in the RL circuit at that frequency.

PARALLEL RC AND RL CIRCUITS

Experiment # 7

Introduction

The experiment on Parallel RC and RL Circuits is a vital educational undertaking designed to deepen the understanding of circuit dynamics when resistors are combined with capacitors (RC circuits) and inductors (RL circuits) in a parallel configuration. Unlike series circuits, parallel configurations offer unique challenges and insights, particularly in how components share voltage and contribute to the overall impedance of the circuit. This is crucial in the realms of electronics and electrical engineering, where parallel circuits are commonly encountered in filter design, power distribution, and signal processing.

In this experiment, participants will construct, examine, and analyze the behavior of RC and RL circuits arranged in parallel. The focus will be on understanding the individual and combined behaviors of capacitors and inductors when they are connected parallel to resistors. This setup provides a rich ground for exploring a range of dynamic electrical phenomena.

Learning Objectives:

1. **Comprehension of Parallel Circuit Principles:** Participants will learn the fundamental differences between series and parallel circuits, especially how voltage, current, and impedance behave in parallel configurations.
2. **Construction and Analysis of Parallel RC Circuits:** Participants will build parallel RC circuits and study their characteristics, focusing on concepts such as capacitive reactance, time constant, and phase angle.
3. **Exploring Parallel RL Circuits:** Similarly, participants will construct parallel RL circuits, observing the behavior of inductive reactance, time constant, and the phase relationship between voltage and current.
4. **Understanding Impedance in Parallel Circuits:** A key objective is to understand how impedance is calculated in parallel RC and RL circuits and how it affects the circuit's response to AC signals.
5. **Frequency Response Analysis:** Participants will explore how the frequency of the input signal affects the behavior of parallel RC and RL circuits, which is fundamental in applications like filtering and tuning.
6. **Use of Measurement Tools:** Practical skills in using tools like oscilloscopes, function generators, and multimeters to observe and measure circuit parameters will be emphasized.
7. **Real-World Applications:** The experiment will discuss the practical applications of parallel RC and RL circuits in various electronic and electrical systems.
8. **Critical Thinking and Problem-Solving:** Encouraging analytical thinking and problem-solving by having participants interpret the results and understand the implications of changes in circuit parameters.

Upon completion of this experiment, participants should be able to understand the dynamics of parallel RC and RL circuits, how to construct and analyze them, and appreciate their significance in various electronic applications. This experiment is vital for those seeking to deepen their knowledge in electronics, especially in areas involving circuit analysis and design.

PRE-LAB:

Reading:

3. Study the Background section of this Laboratory experiment.

Written:

4. In your lab notebook sketch the circuit diagram to be used in the procedure.
5. Prepare tables to record the data.

MATERIALS REQUIRED

Power Supply:

- Power source for oscilloscope

Instruments:

- Oscilloscope
- DMM
- Function Generator

Miscellaneous:

- Assorted leads for connecting to the oscilloscope terminals and jacks
- 1 - 10 k Ω
- 1 - 2.2 k Ω
- 2 - 100 Ω (discrete)
- 2 - 22 Ω (discrete)
- 1 - 100 mH
- 1 - 0.01 μ f

*Components may be discrete or via decade substitution boxes

BACKGROUND

As was seen in prior experiments, in a series circuit the same *current* is in all components, and so *current* is generally used as a reference in series circuits. However, in parallel circuits, the same *voltage* is across all components, so *voltage* is the logical and appropriate reference. The current in each branch then evolves from the circuit voltage.

For series RC and RL circuits, we saw that Kirchhoff's voltage law applies, but that the voltages must be added as phasors. Similarly, in parallel circuits, Kirchhoff's current law applies to any junction, but again, the currents must be added as phasors. The current entering a junction is always equal to the current leaving the junction.

In a parallel circuit, if the impedance of each branch is known, then the current in that branch can be determined directly from the applied voltage and Ohm's law. The current phasor diagram can then be constructed, and the total current can be found as the phasor sum of the currents in each branch.

Consider the current phasor diagram for the parallel RC circuit shown in Figure 5.1. The current in the capacitor is shown at +90° from the voltage reference because the current leads the voltage in a

capacitor. The current in the resistor is along the x-axis because current and voltage are in phase in a resistor.

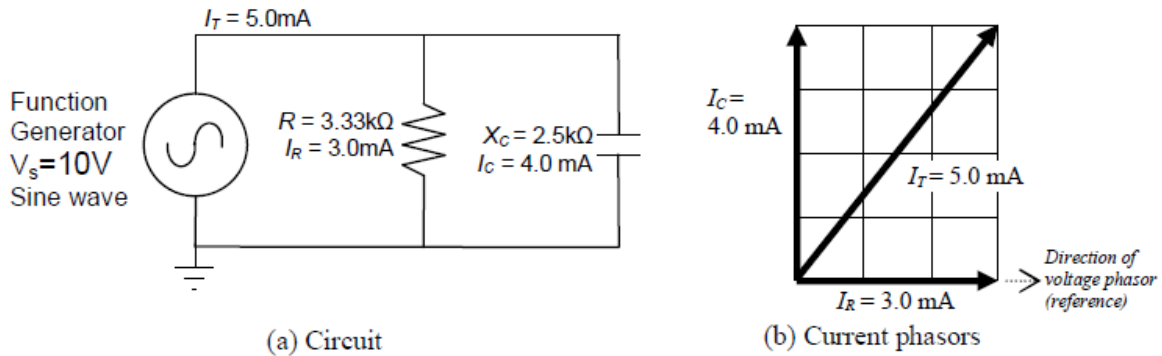


Figure 5.1 Parallel RC Circuit

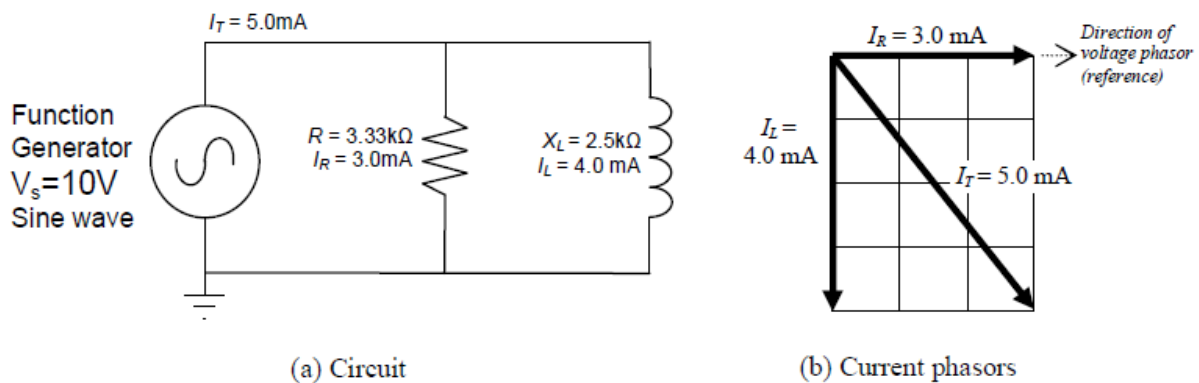


Figure 5.2 Parallel RL Circuit

Figure 5.2 shows a parallel RL circuit and its current phasor diagram. Here we have assumed all components are "ideal". The current in an ideal inductor is at -90° from the voltage reference, because the current lags the voltage in an inductor. However, practical inductors contain resistance that often is large enough to affect the phasor. The effect of the inductor's resistance on the phasor diagram would be to reduce the angle between the I_L and I_R . In a practical circuit, this angle will be slightly less than the -90° of a pure inductor. This experiment will illustrate this difference between the approximation of circuit performance based on ideal components and the actual measured values.

For both RC and RL circuits, the Pythagorean theorem and ordinary vector addition can be applied to the current phasors to determine the magnitude of the total current, I_T .

$$I_T = \sqrt{I_R^2 + I_C^2} \text{ and } I_T = \sqrt{I_R^2 + I_L^2}$$

Recall that in series circuits, the phase angle was measured between the source voltage V_s and the resistor voltage V_R using the oscilloscope. The oscilloscope is a voltage-sensitive device, so examining those voltages and phase angles is straightforward. But in parallel circuits, the phase angle of interest usually is between the total current, I_T , and one of the branch currents. To use the oscilloscope to measure the phase angle in a parallel circuit, we must convert the current to a voltage. This is commonly done by inserting a small resistor (a "sense" resistor) in the branch with the current to be measured. Such a sense

resistor makes it easy to determine the magnitude and the phase of the current in that branch, but the resistor must be small enough not to have a significant effect on the values to be measured.

Parallel RC Circuit

This part of the experiment will give you experience making the measurements with the digital multimeter (DMM). Because the electrodes of this device are isolated from the circuit ground and the grounds in NI-ELVIS, you may make voltage measurements directly across any of the components, whether or not they are grounded.

1. Create a table similar to Table 5.1 for recording experimental data: R_1 , R_{S1} , R_{S2} , C , f , V_s , V_{RS1} , V_{RS2} , V_{R1} , I_T , I_C , I_{R1} .

Table 5.1

	Nominal Value	Measured Value	Voltage Across R		Current Calculated	
R_{S1}	100 Ω			V_{RS1}		I_T
R_{S2}	100 Ω			V_{RS2}		I_C
R_1	100 Ω			V_{R1}		I_{R1}
C	0.01 μf					I_C
V_s	2.0 Vrms					
f	1.0 kHz					

2. Measure circuit components using the DMM: one 10 k Ω resistor, two 100 Ω sense resistors, and one 0.01 μF capacitor. (Be sure to wait until the capacitance measurement stabilizes.) Record the measured values in Table 5.1.
3. Construct the circuit shown in Figure 5.3. Set the function generator to provide a sine wave with a voltage of 2.0 Vrms at 1.0 kHz. Verify the voltage and frequency with your oscilloscope while the circuit is connected and operating; adjust if necessary.

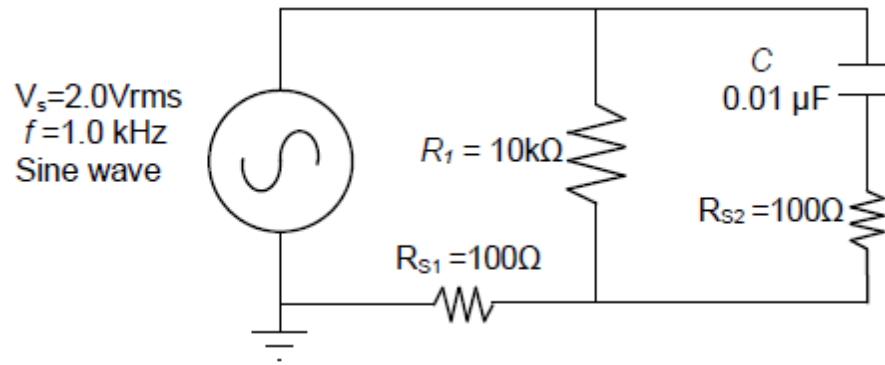


Figure 5.3

4. Using the DMM voltmeter in AC mode [V~], measure the voltage drop across each resistor. The voltage drops are small, so measure as accurately as possible and keep three significant figures in your measurement. Record the voltage drops in Table 5.1.
5. Compute the current in each resistor using Ohm's law, and record the calculated currents in Table 5.1.
6. Draw the current phasors I_{R1} , I_C , and the total current I_T in a plot similar to that of Figure 5.1. The total current is through sense resistor R_{s1} . The current I_C is through sense resistor R_{s2} . Ignore the small effect of the sense resistors on the phasor diagram. Note carefully the direction of the phasors. Label each of the current phasors.
7. Compute X_C for the 1.0 kHz frequency and record the value in Table 5.2. Using this value and that of the sense resistor, calculate the expected current, I_C , through the capacitor. Record the value in the available space in Table 5.1 on the line listing the capacitor's value. How does this value compare to that found from the sense resistor?
8. Using the value of X_C for the 1.0 kHz frequency and the measured resistance of R_1 , find the total impedance, Z_T , of the circuit. Remember that these impedances add like parallel resistors. Ignore the sense resistors for this calculation. Show your work.
9. Using Z_T and the applied voltage, V_s , compute the total current, I_T . Show your work. The total current should reasonably agree with the value determined in step 5.

Table 5.2

X_C	Z_T	V_s	$I_{T_computed}$

10. Change the frequency to 2.0 kHz at 2.0 V_{rms} . Repeat steps 4, 5, and 6, recording the results in a separate table.

Parallel RL Circuit

This part of the experiment will give you experience making similar measurements using the oscilloscope. Remember that the ground lead of the oscilloscope must be connected to circuit ground, that is, the ground of the function generator, in this case. **Do not try to make measurements directly across ungrounded components.** Also, in this part of the experiment, record all voltages and currents as *peak-to-peak* values.

1. Create a table similar to Table 5.3 for recording experimental data: R_1 , R_{S1} , R_{S2} , L , f , V_S , V_{RS1} , V_{RS2} , V_{R1} , I_T , I_L , I_{R1} .

Table 5.3

	Nominal Value	Measured Value	Voltage Across R		Current Calculated	
R_{S1}	22 Ω			V_{RS1}		I_T
R_{S2}	22 Ω			V_{RS2}		I_C
R_1	2.2 k Ω			V_{R1}		I_{R1}
L	100 mH					I_L
V_S	4.0 V_{pp}					
f	4.0 kHz					

2. Using the DMM, measure the actual values of a 2.2 k Ω resistor, two 22 ohm resistors, and a 100 mH inductor.
3. Measure the winding resistance of the inductor, R_W , with an ohmmeter.

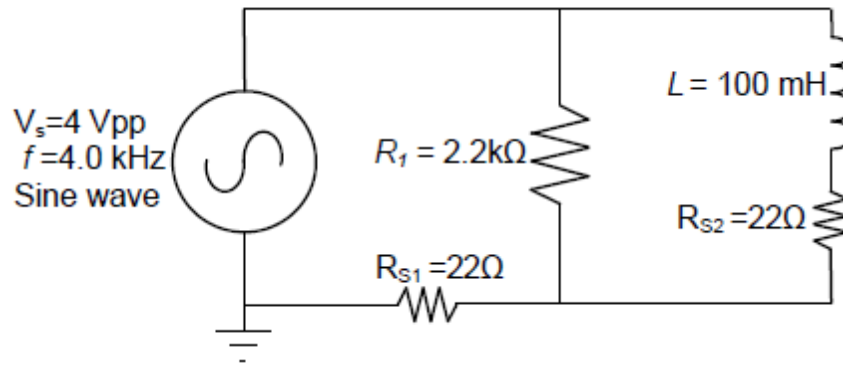


Figure 5.4

4. Construct the circuit shown in Figure 5.4. Using the oscilloscope's CHANNEL 0 to monitor the function generator, set the source voltage to a sine wave with a voltage of 4.0 V_{PP} at 4.0 kHz. Verify both the voltage and the frequency with the oscilloscope.
5. Using oscilloscope CHANNEL 1, measure the peak-to-peak voltage across R_{S1}. Apply Ohm's law to calculate the peak-to-peak value of the total current, I_T.
6. Because R_{S1} is small compared to R₁, the source voltage V_S is very nearly the voltage across the resistor R₁. Furthermore, since only resistors are in that loop, the phase of the voltage across R₁ — and current through R₁ — is the same as the phase of the source voltage V_S.

With CHANNEL 0 of the oscilloscope displaying the voltage across the function generator (V_S), and CHANNEL 1 displaying the voltage across R_{S1}, measure the phase angle between the generator voltage, V_S, and the generator current, I_T, (i.e., the current flowing through R_{S1}). This is equivalently the phase angle between I_{R1} and I_T.

To make the measurement, set the scope's TRIGGER to EDGE and trigger on CHANNEL 0. Use the AUTOSCALE button to ensure the waveforms are approximately the same height. You may press STOP to improve accuracy when taking cursor measurements. Record the measured phase angle between I_{R1} and I_T.

7. Replace the sense resistor R_{S1} with a jumper wire. CHANNEL 0 is still measuring the output voltage of the function generator, which now is also the voltage across R₁. Connect CHANNEL 1 of the oscilloscope across sense resistor R_{S2}. Record the voltages across R₁ and across R_{S2}. Apply Ohm's law to calculate the current in each branch of the circuit, and record the currents. Since L is in series with R_{S2}, enter the same current for both in Table 5.3.
8. Using the computed peak-to-peak currents from Table 5.3, draw the current phasor diagram for the circuit. Ignore the effects of the sense resistors. Your diagram should look somewhat similar to Figure 5.2 (b).
9. The phasor diagram depicts visually the relationship between the total current and the currents in each branch of the circuit. From the currents in the phasor diagram, compute the phase angle between the total current (I_T) and the current in R₁ (I_{R1}). Then compute the phase angle between the total current (I_T), and the current in L (I_L). Ideally, what should be the sum of these angles?

10. On the oscilloscope, measure the angle between I_L and I_{R1} . (The oscilloscope's leads were already connected to do this as a result of step 7.) Ideally, this measurement should be 90° , but resistance in the inductor, may reduce the angle. If necessary, adjust both scope channels to have the same apparent height on the oscilloscope screen in order to make the measurement.
11. In step 6 you measured the phase angle Φ_{RT} between I_{R1} and I_T . In step 10 you measured the phase angle Φ_{RL} between I_{R1} and I_L . Compute the phase angle Φ_{TL} between I_T and I_L by subtracting Φ_{RT} from Φ_{RL} . That is, $\Phi_{TL} = \Phi_{RL} - \Phi_{RT}$
12. Construct a table that will allow easy comparison of the computed and measured phase angles from steps 6, 9, 10, and 11. Compare the measured phase angles versus the computed phase angles. Discuss likely causes for any discrepancies.

POWER IN AC CIRCUITS

Experiment # 8

Introduction

The experiment on Power in AC (Alternating Current) Circuits is an essential educational initiative designed to enhance understanding of how power is managed, calculated, and utilized in AC electrical systems. AC power is fundamental to the functioning of modern electrical grids and a wide array of electronic devices. This experiment is crucial for students and professionals in electrical engineering, electronics, and related fields, as it provides practical insights into the complexities and nuances of power in AC circuits, which differ significantly from those in direct current (DC) systems.

In this experiment, participants will explore various aspects of power in AC circuits, including real power, reactive power, apparent power, and power factor. The objective is to understand not only how to measure these different forms of power but also their significance and impact on electrical systems.

Learning Objectives:

Understanding AC Power Concepts: Participants will learn the fundamental principles of AC power, including the differences between real, reactive, and apparent power.

Measurement of Power Parameters: The experiment will cover methods and techniques to measure real power (watts), reactive power (vars), and apparent power (volt-amperes) in AC circuits.

Exploring Power Factor: One of the key learning aspects will be understanding the concept of power factor, its calculation, and its significance in AC circuits. This includes exploring how a non-ideal power factor can affect efficiency and energy costs.

Analyzing AC Circuit Behavior: Participants will observe and analyze how various components like resistors, inductors, and capacitors affect power consumption and the power factor in AC circuits.

Use of Instrumentation: Practical skills in using tools like power meters, oscilloscopes, and multimeters to measure and analyze AC power parameters.

Real-World Implications: Understanding the implications of AC power characteristics in real-world applications, such as in power transmission, motor operation, and energy-efficient system design.

Problem-Solving and Critical Analysis: Encouraging participants to apply critical thinking and problem-solving skills in analyzing and interpreting the behavior of AC circuits concerning power.

Upon completion of this experiment, participants should have a comprehensive understanding of the various dimensions of power in AC circuits, including how to measure and analyze them, and the practical implications of these measurements. This knowledge is crucial for anyone involved in the design, operation, or analysis of electrical systems, particularly those that rely on alternating current.

Apparent power is the input power to the ac circuit. It is defined as the product of the voltage V and the current I in the AC circuit. Thus,

$$P_A = V \times I$$

for lack of better unit and to avoid confusion with the unit watts, apparent power is usually measured in voltamperes (VA).

Current through a reactance produces reactive power

$$P_X = V_X \times I$$

which is usually abbreviated VAR.

The power in watts consumed by an electrical device with both resistive and reactive components is defined as true power. In a circuit containing both resistive and reactive components, true power is equal or less than apparent power. The amount by which apparent power must be multiplied to obtain true power is called the power factor (pf) of the circuit. It is really the ratio of the true power to apparent power. Thus,

$$pf = \frac{\text{true power}}{\text{apparent power}} = \frac{P(W)}{P_A(VA)}$$

MATERIALS REQUIRED

Power Supply:

- Variable AC Power supply

Instruments:

- DMM

Miscellaneous:

- 1 - 100-Ω, 5-W
- 1 - 5 uF or 4.7 uF
- 1 - 10 μF, 100 V

The Voltage-Current Method

1. Refer to Figure 8.1 for circuit connections. Using an ohmmeter, measure the resistance of the 100 Ω resistor, and record the value in Table 8.1.
2. Increase the output voltage of the variable ac power supply until it $V_{AB} = 50$ V.
3. Measure the voltage across the resistor V_R and the current I . Record the values in Table 8.1.

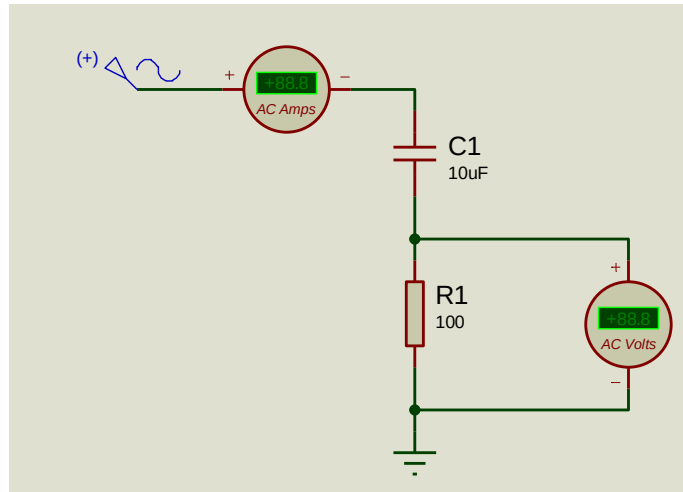


Figure 8.1 Circuit Connections

4. Turn of the power supply.
5. Calculate the apparent power P_A , the true power P , the power factor, and the phase angle of the circuit. Use the measured values of V_{AB} , V_R , and I in your calculations. Record your answers in Table 8.1
6. With the power supply turned off replace the 5 uF with a 10 uF capacitor.
7. Increase the output of the power supply until $V_{AB} = 25$ V.
8. Measure the voltage across the resistor V_R and the current I . Record the values in Table 8.1.
9. Repeat step 5 for 10 uF capacitor. Record your answers in Table 8.1.

Resistance R, Ω		Capacitance (Rated Value) C, uF	Applied Voltage V_{AB} , V	Voltage across Resistor V_R , V	Current (Measured) I, mA	Apparent Power P_A , VA	True Power P, W	Power Factor pf	Phase Angle θ , degrees
Rated Value	Measured Value								
100		5							
100		10							

Table 8.1 Power Measurement

Experiment 9

FIRST ORDER R-L AND R-C CIRCUITS

The study of first-order R-L (Resistor-Inductor) and R-C (Resistor-Capacitor) circuits is fundamental in understanding the behavior of electric circuits involving energy storage elements. These circuits are called first-order because their behavior is described by first-order differential equations. In these circuits, the resistor (R) provides resistance to the flow of electric current, while the inductor (L) and capacitor (C) are the energy storage elements. Inductors store energy in a magnetic field when current flows through them, and capacitors store energy in an electric field as a charge on their plates.

The behavior of these circuits is significantly influenced by the time constant, denoted as τ . For an R-L circuit, the time constant is $\tau=L/R$, where L is the inductance and R is the resistance. For an R-C circuit, the time constant is $\tau=RC$, where R is the resistance and C is the capacitance. This time constant determines how quickly the circuit responds to changes in voltage or current.

Understanding these circuits involves analyzing how voltage and current change over time, especially during transient conditions – the periods after a circuit is switched on or off, before reaching a steady state.

Learning Objectives

1. Understand Basic Concepts: Gain a fundamental understanding of first-order R-L and R-C circuits, including the roles of resistors, inductors, and capacitors.
2. Study of Transient Response: Learn about the transient response in R-L and R-C circuits, focusing on how these circuits behave immediately after a change in voltage or current.
3. Analyze Time Constant: Comprehend the concept of the time constant (τ) and its significance in determining the rate at which current and voltage change in the circuit.
4. Learn to Derive and Solve Differential Equations: Develop the ability to derive the first-order differential equations that govern the behavior of R-L and R-C circuits and solve them for various input conditions.
5. Circuit Analysis Techniques: Master various techniques for analyzing these circuits, including the use of Kirchhoff's laws, Thevenin's theorem, and Norton's theorem.
6. Practical Application and Experimentation: Engage in practical experiments to observe the transient behavior of R-L and R-C circuits, measure time constants, and compare theoretical predictions with experimental data.
7. Interpretation of Results: Enhance skills in interpreting and analyzing the results of these experiments, including understanding the effects of varying circuit parameters.
8. Understanding Steady-State Conditions: Learn about steady-state conditions in R-L and R-C circuits and how they differ from transient states.

By mastering these objectives, students will gain a comprehensive understanding of first-order R-L and R-C circuits, which are crucial in various fields of electrical engineering and electronics.

I. BACKGROUND

I.1 Measurement of the Natural Response of First Order Circuits.

The natural response of an R-C circuit, shown in Figure A, is given by (1).

$$v(t) = v_o \cdot e^{-\frac{t}{\tau}}$$

Where, v_o is the value of the capacitor voltage v at $t=0$ s, $\tau=RC$ is the time constant of the circuit. The time constant gives the rate at which the voltage decays to zero. In circuits, this decay response is due to ohmic losses.

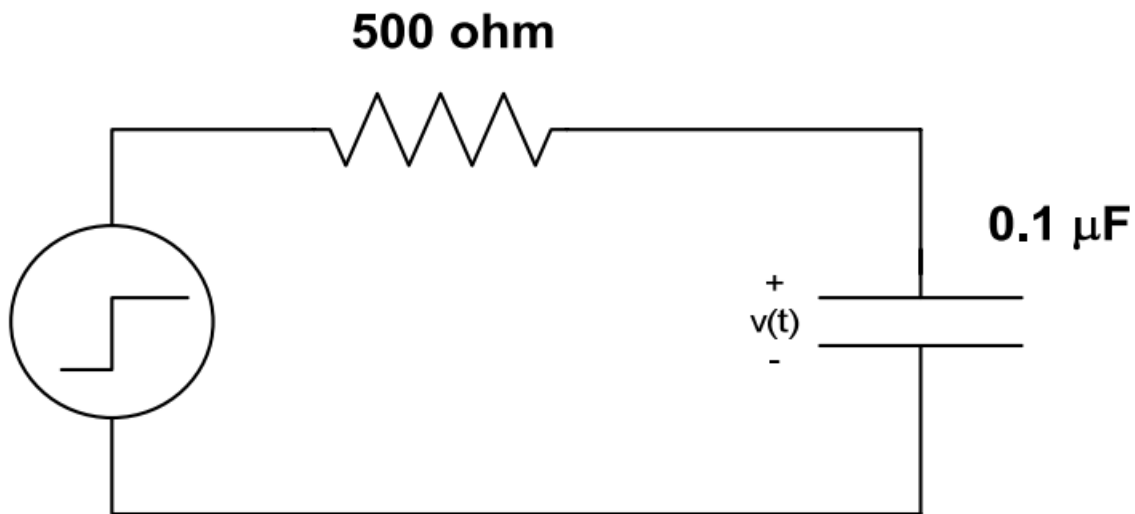


Figure A

The time constant can be measured by one of the following graphical methods.

(a) The Tangential Line. A line tangential at a certain point of the response curve is drawn. The line intersects the time axis exactly in one time constant from the point of tangent. Figure B shows the graph of the response from (1). Note that the response begins at $t=0.05$ s. With reference to Figure B, a line is drawn tangential to the curve at $t=0.05$ s (that is at the onset of the decay). The line is extended until it intersects the time axis. This occurs at $t=0.25$ s. Therefore, the time constant of the response is

$$\tau = 0.25 - 0.05 = 0.2 \text{ s (2)}$$

This method is suitable if a hard copy of the response graph is available.

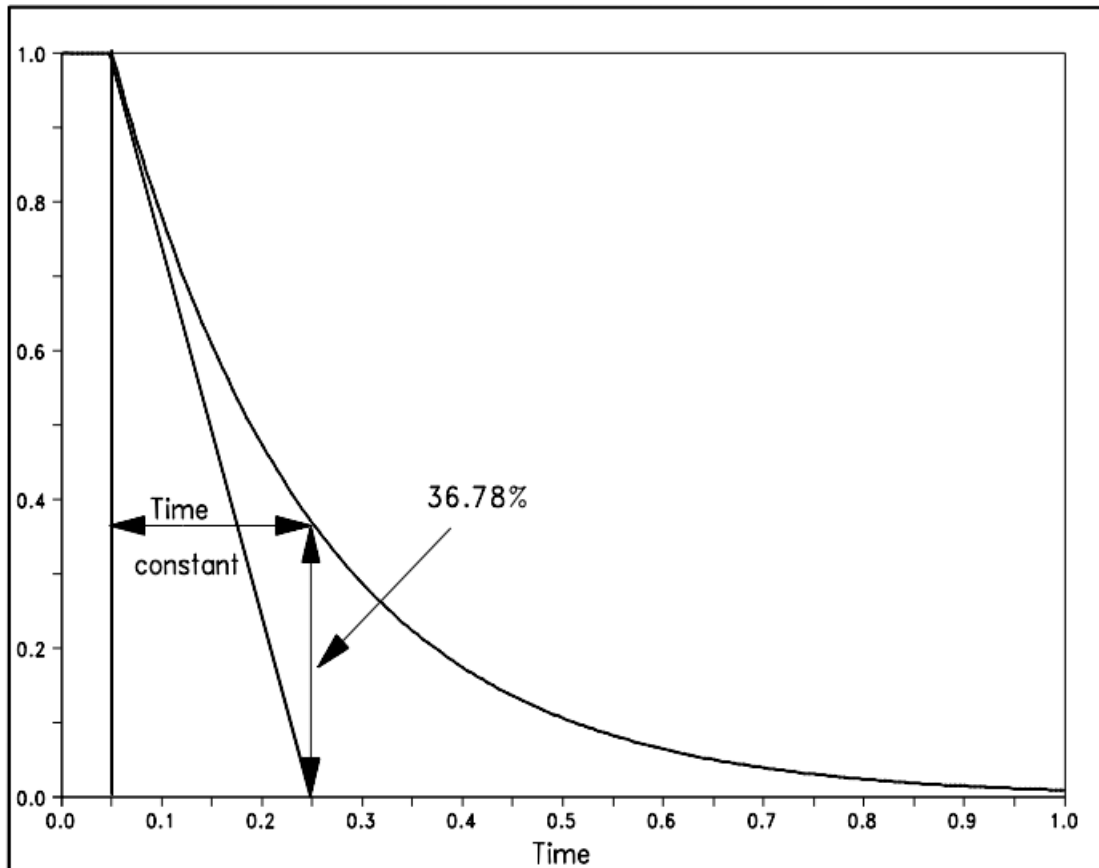


Fig. B. Measurement of the time constant of an exponentially decaying response.

(b) The 63.22% Decay. The second method is suitable for measurements on the oscilloscope. From (1), every time interval equal to one time constant the response decays by 63.22%. Equivalently, at the end of every interval equal to one time constant, the response is at the $100 - 63.22 = 36.78\%$ of its value at the beginning of the interval. This method of measurement is demonstrated in Figure B. The response begins its decay at $t=0.05$ s, at that point the value of the curve is 1. The response reaches 36.78 % of its initial value at $t=0.25$ s. Thus the time constant is

$$t = 0.25 - 0.05 = 0.2 \text{ s} \quad (3)$$

The results of (3) and (2) are in agreement.

1.2 The Step Response of First Order Circuits.

The step response of the first order circuit in Figure A is given by (4).

(t)

$$v(t) = v_f \left(1 - e^{-\frac{t}{\tau}} \right)$$

(4)

Where, v_f is the final value of the response (also, called the constant steady state of the response). The

step response of the circuit is characterized by the same time constant as the natural response. The time constant affects the step response in a similar manner as the natural response. Therefore, the methods discussed previously apply in this case, as well.

On the oscilloscope the time constant of the step response can be measured by measuring the time the output requires to reach 63.22% of its final value.

The natural and step responses of a first order R-L circuit, shown in Figure C, are the same as in (1) and (4) given for the circuit current i . The circuit time constant is $\tau=L/R$.

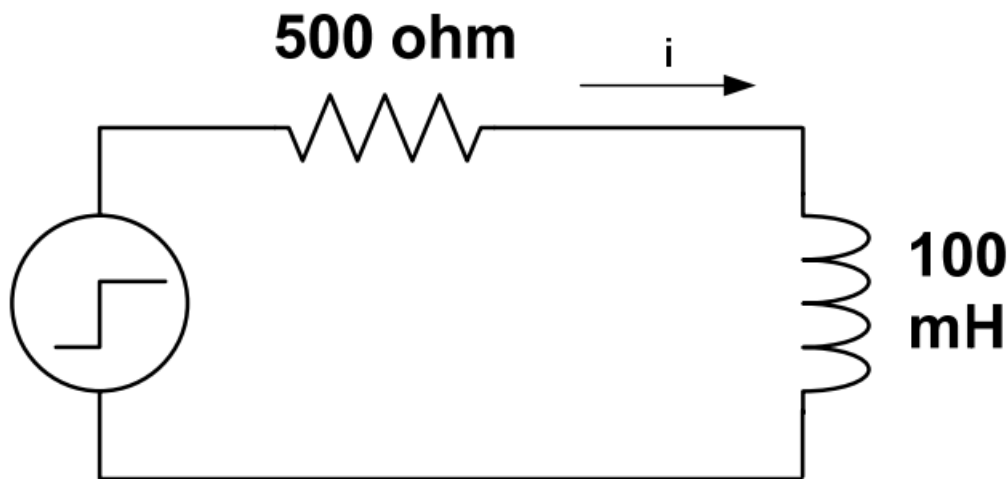


Fig. C. The experimental R-L circuit.

I.3 Oscilloscope Observations.

Non-storage oscilloscopes must display periodic wave forms, so that measurements and observations can be easily made. The natural response of a circuit can be observed along with its step response, if a square-pulse periodic excitation is applied to the circuit.

Figure D shows the response of a first order circuit to a periodic square pulse. The first part of the circuit response is the step response described by (4). The second part of the circuit response, when the excitation becomes zero, is the natural response described by (1). If the response is allowed to decay sufficiently, the following period will begin with an other step response followed by the natural response. Thus, the circuit response will alternate between step and natural responses. To achieve good results, therefore, the period of the excitation must be adjusted to 7 to 10 times the time constant of the circuit.

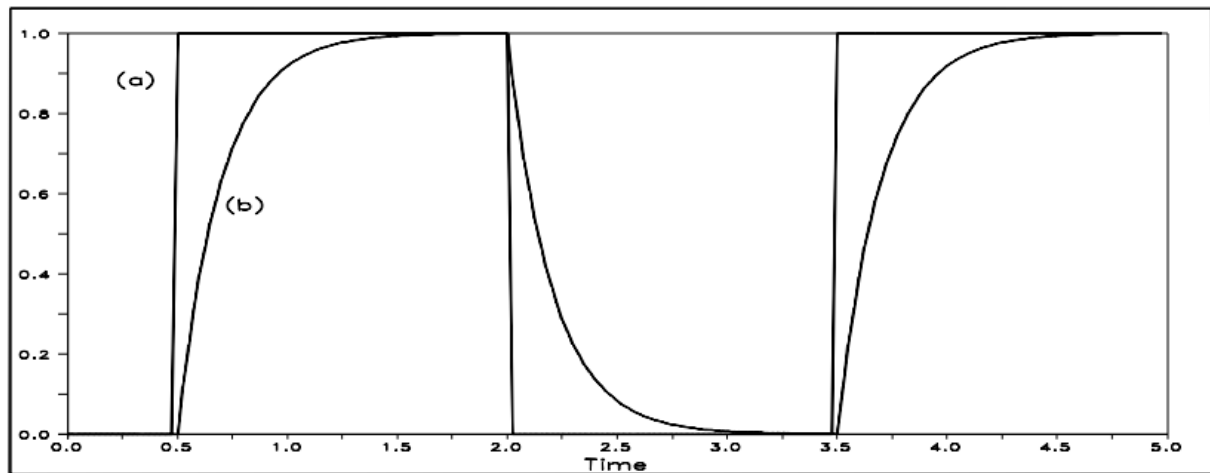


Fig. D. Circuit response to a square pulse excitation with large period. (a) The excitation. (b) The circuit response.

II. INSTRUMENTATION

Function generator, Tektronix CFG 250. Oscilloscope, Tektronix 2205-20 MHz.

III. PROCEDURE:

Become familiar with the signal generator and the oscilloscope.

A. The R-C Circuit.

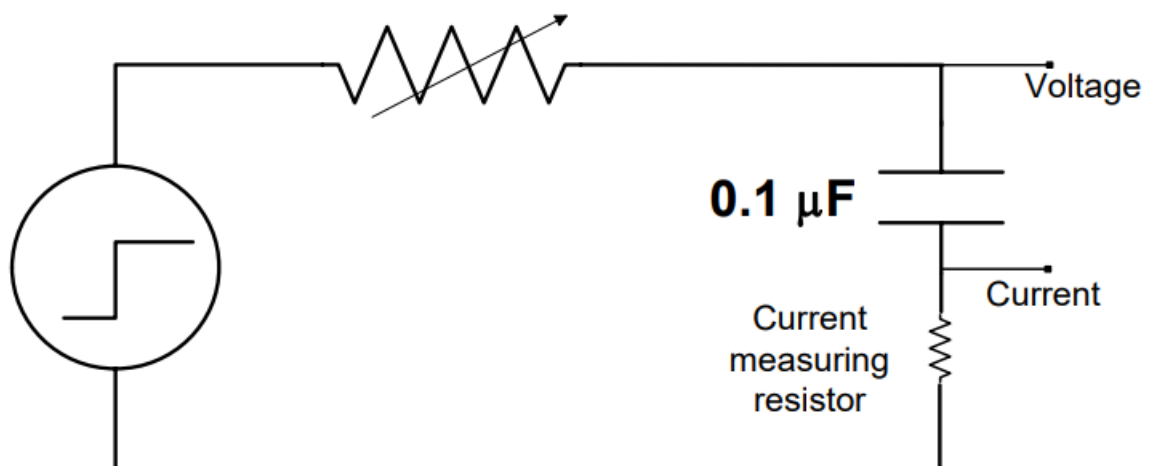


Fig. 1. The experimental R-C circuit.

1. (a) Construct the R-C circuit of Figure 1. Use a 510 Ω .

(b) Apply the square pulse generator to produce a periodic response of the circuit that appears clearly on the oscilloscope. The period of the source is adjusted to be at least 20 times higher than the calculated time constant of the circuit. This is necessary, so that the capacitor reaches steady state, in

both charging and discharging intervals.

(c) Observe on the oscilloscope the quantities shown in the figure. Measure the circuit time constant by the 63% decay method described above.

(d) Plot on a millimeter paper one period of the voltage and current response. Use appropriate scale. Label the axes. Transfer measurements from the oscilloscope to the paper by inspection.

2. Change the resistance in Figure 3 to 250 Ω . Repeat Procedure 1. Use the same millimeter paper and the same scale. Align the graphs. Do not change the setting of the source.

B. The R-L Circuit.

3. (a) Construct the circuit of Figure 2. Set the resistance box to 500 Ω .

(b) Apply a square pulse voltage. Set the frequency of the source to correspond to 10 times the time constant of the circuit.

(c) Repeat the same step 1c for this circuit.

4. Change the resistance to 250 Ω . Perform again step 1c.

5. Simulate the experiment circuit (Figure 2) in Pspice. For each value of R used above compute the circuit time constant.

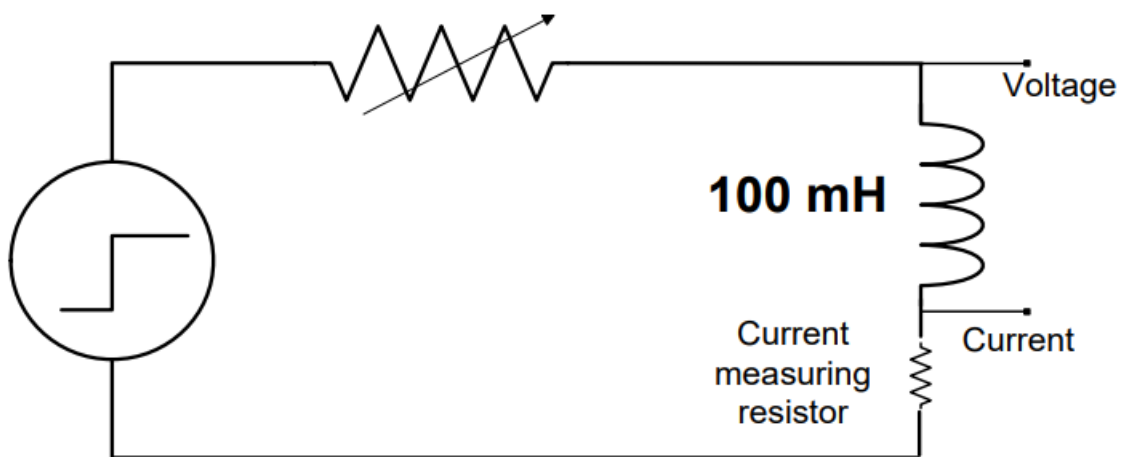


Fig. 2. The experimental R-L circuit.

IV. REPORT

A. Theoretical Development.

1. In theory discuss the natural and step responses of R-C and R-L circuits.

2. Define the duty cycle of a pulse. What is the duty cycle of a square pulse?

3. What is the difference in the setting of x1 and x10 of the oscilloscope probes?

4. How do we adjust the voltage and time scale of the oscilloscope? What is the purpose of calibrating a channel?

5. What is a waveform? What kind of waveforms are best observed and measured by oscilloscopes similar to the one used in the experiment?

6. What is a current measuring resistor? What was the value of the current measuring resistor used in the experiment? Is this value satisfactory?

B. The R-C Circuit.

7. (a) Explain the response observed in Procedure 1. Indicate which interval represents the natural and which interval represents the step response of the circuit.

(b) What is the circuit time constant measured for each value of R?

C. Pspice Simulation of the R-L Circuit.

8. Compare the time constant values obtained from the simulations with those obtained from the measurements.

EXPERIMENT 10:

SECOND ORDER RLC CIRCUITS

Introduction

The exploration of second-order RLC (Resistor-Inductor-Capacitor) circuits forms a critical part of advanced studies in electrical engineering. Unlike first-order circuits, which contain only one reactive element (either an inductor or a capacitor), second-order RLC circuits include both an inductor and a capacitor. This combination results in more complex behavior characterized by second-order differential equations.

In RLC circuits, the resistor (R) provides resistance to the flow of electric current, the inductor (L) stores energy in a magnetic field, and the capacitor (C) stores energy in an electric field. The interplay between these components creates interesting phenomena such as resonance, where the circuit can oscillate at a particular frequency known as the resonant frequency.

The study of second-order RLC circuits is essential for understanding the behavior of many electrical systems, such as radio transmitters and receivers, filters in electronic equipment, and in the study of oscillations in general electrical engineering applications.

Learning Objectives

1. **Understand Basic Components and Concepts:** Gain a thorough understanding of the roles of resistors, inductors, and capacitors in RLC circuits and how they interact with each other.
2. **Analyze Second-Order Differential Equations:** Develop the skill to derive and solve the second-order differential equations that describe the behavior of RLC circuits.
3. **Study Resonance Phenomena:** Learn about the resonance in RLC circuits, including how to calculate the resonant frequency and understand its significance in practical applications.
4. **Examine Transient and Steady-State Responses:** Understand the transient (temporary) and steady-state (long-term) responses of RLC circuits, especially how these circuits behave in response to different types of inputs, such as step, sinusoidal, and impulse inputs.
5. **Explore Damping and Q-Factor:** Investigate the concepts of damping and the quality factor (Q-factor) in RLC circuits, which affect how quickly the transient response dies out and the sharpness of the resonance peak.
6. **Practical Application and Experimentation:** Conduct experiments to observe the behavior of RLC circuits in real-world conditions, measure their resonant frequency, and analyze how circuit parameters affect their behavior.
7. **Circuit Analysis Techniques:** Master various techniques for analyzing these circuits, including using complex impedance, phasor diagrams, and network theorems like Thevenin's and Norton's theorems.
8. **Interpretation of Experimental Data:** Develop skills in interpreting experimental data, comparing it with theoretical predictions, and understanding the effects of real-world non-idealities on

circuit behavior.

9. Applications in Real-World Systems: Understand how RLC circuits are used in practical applications, such as in filters, oscillators, and tuning circuits in communication systems.

By achieving these objectives, students will be well-equipped with a comprehensive understanding of second-order RLC circuits, which are pivotal in various advanced applications in electrical and electronic engineering.

I. BACKGROUND

I.1 The Natural Response of a Second Order Circuit.

The natural response of second order circuit is derived from the solution of its characteristic differential equation (1).

$$\frac{d^2 y}{dt^2} + 2 \cdot \beta \cdot \frac{dy}{dt} + \omega_o^2 \cdot y(t) = 0$$

In (1) above, β is the damping coefficient and ω_o is the frequency of undamped oscillations of the circuit (also called the resonance frequency for under damped circuits). The type of response derived from (1) is determined by the above two parameters. We distinguish these cases:

- (a) $\beta < \omega_o$. The response is under damped. The voltage and current oscillate with a diminishing amplitude.
- (b) $\beta = \omega_o$. The response is critically damped.
- (c) $\beta > \omega_o$. The response is over damped. The voltage and current decay exponentially.

The under damped response is characteristic of second order circuits. Hence, we emphasize it.

I.2 Measurement of the Natural Under damped Response.

The natural response of an under damped circuit is given by (2).

$$y(t) = Y_m \cdot e^{-\beta t} \cdot \cos(\omega_d \cdot t + \phi) \quad (2)$$

Where, ω_d is the frequency of damped oscillations given by (3). The parameters Y_m and ϕ are found from the initial conditions of the circuit according to (4).

$$\omega_d = \sqrt{\omega_o^2 - \beta^2}$$

$$y(0) = Y_m \cdot \cos(\phi)$$

$$\left. \frac{dy}{dt} \right|_{t=0^+} = -\beta \cdot Y_m \cdot \cos(\phi) - \omega_d \cdot Y_m \cdot \sin(\phi)$$

Figure A shows an under damped response. The envelope of the response is exponential with a time constant $\tau=1/\beta$.

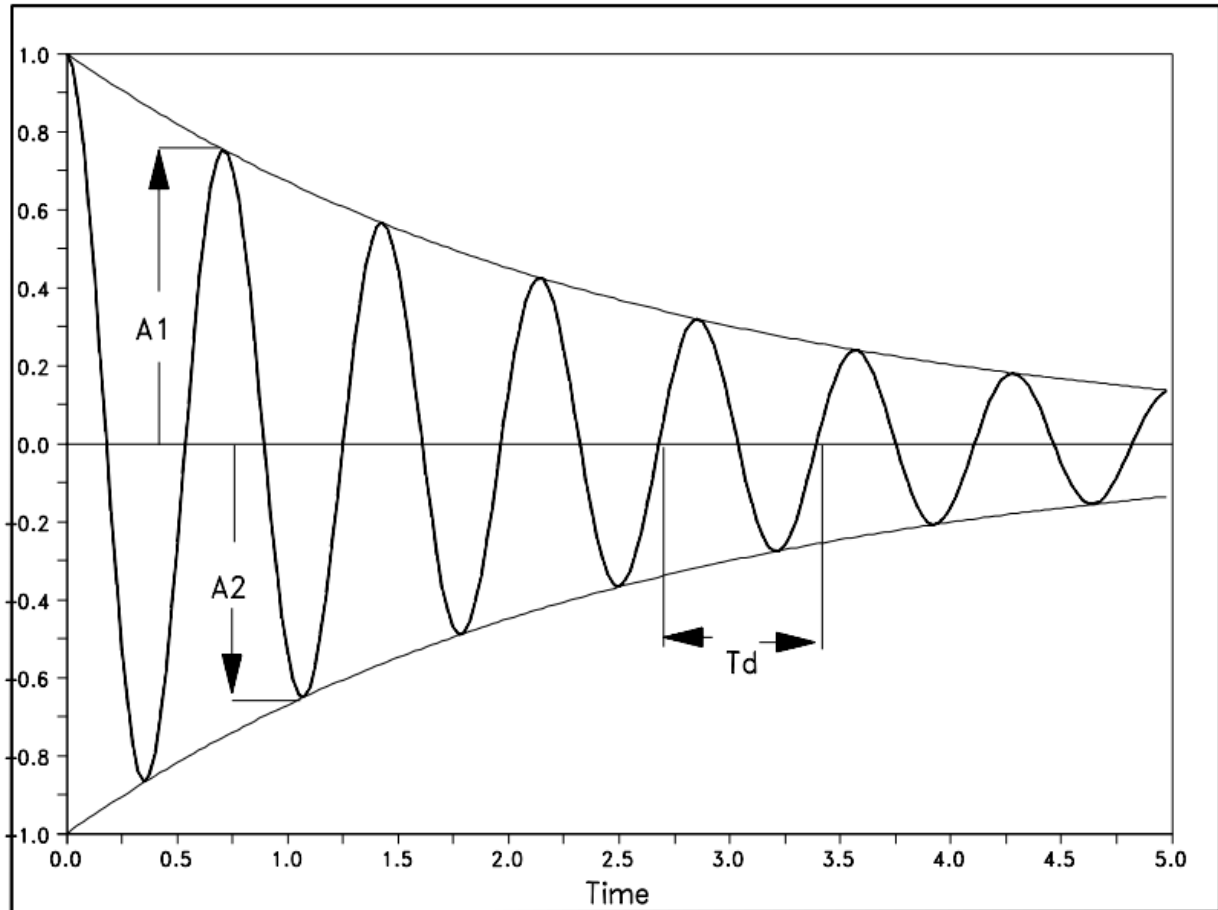


Fig. A. The responses of an under damped second order circuit.

The response can be also expressed in terms of the damping ratio according to (5).

$$y(t) = Y_m \cdot e^{-2\zeta \frac{t}{T_d}} \cdot \cos(\omega_d \cdot t + \phi)$$

Where the damping ratio is

(5)

$$\zeta = \frac{\beta \cdot T_d}{2} = \frac{\pi \cdot \beta}{\omega_d} \quad (6)$$

The damping ratio expresses the ratio between two successive peaks of the oscillation. With reference to Figure A, the ratio between the two successive peaks, A1 and A2, occurring at t_1 and $t_1+T_d/2$ respectively (one half period apart), is, by using (5):

$$\frac{A_2}{A_1} = \frac{e^{-2\zeta \frac{t_1 + T_d/2}{T_d}}}{e^{-2\zeta \frac{t_1}{T_d}}} = e^{-\zeta}$$

The damping ratio shows the speed with which the response decays. For damping ratio equal 1, the amplitude of the oscillation decays by 63.22% in a half cycle (that is between two successive peaks). Thus, the response decays with the same frequency as the frequency of the oscillations. If the damping ratio is 2, the response decays with twice the frequency of the oscillations, and so on. An equivalent way of looking at the damping ratio is this: the inverse of the damping ratio is the number of additional successive peaks it takes for the amplitude of the oscillation to decay by 63.22% from an initial peak. Thus, a response with a damping ratio of 0.5 will decay to 63.22% of an initial amplitude in 2 successive peaks (one full cycle of the oscillation).

I.3 The Step Response of an Under damped Second Order Circuit.

The step response of an under damped second order circuit is shown in Figure B. The same measurements can be performed on this wave form as in the natural response. Three additional parameters of the response are of interest. These are explained in the following with reference to Figure B.

- (a) The response time, T_r , is the time lapsed from the onset of the excitation to the moment the response becomes equal, for the first time, to its steady state value (1 in the figure).
- (b) The overshoot time, $T_{o.s}$, is the time the response reaches its first peak.
- (c) Overshoot error, $Y_{o.s}$, is the amount the response rises in excess of its steady state value.

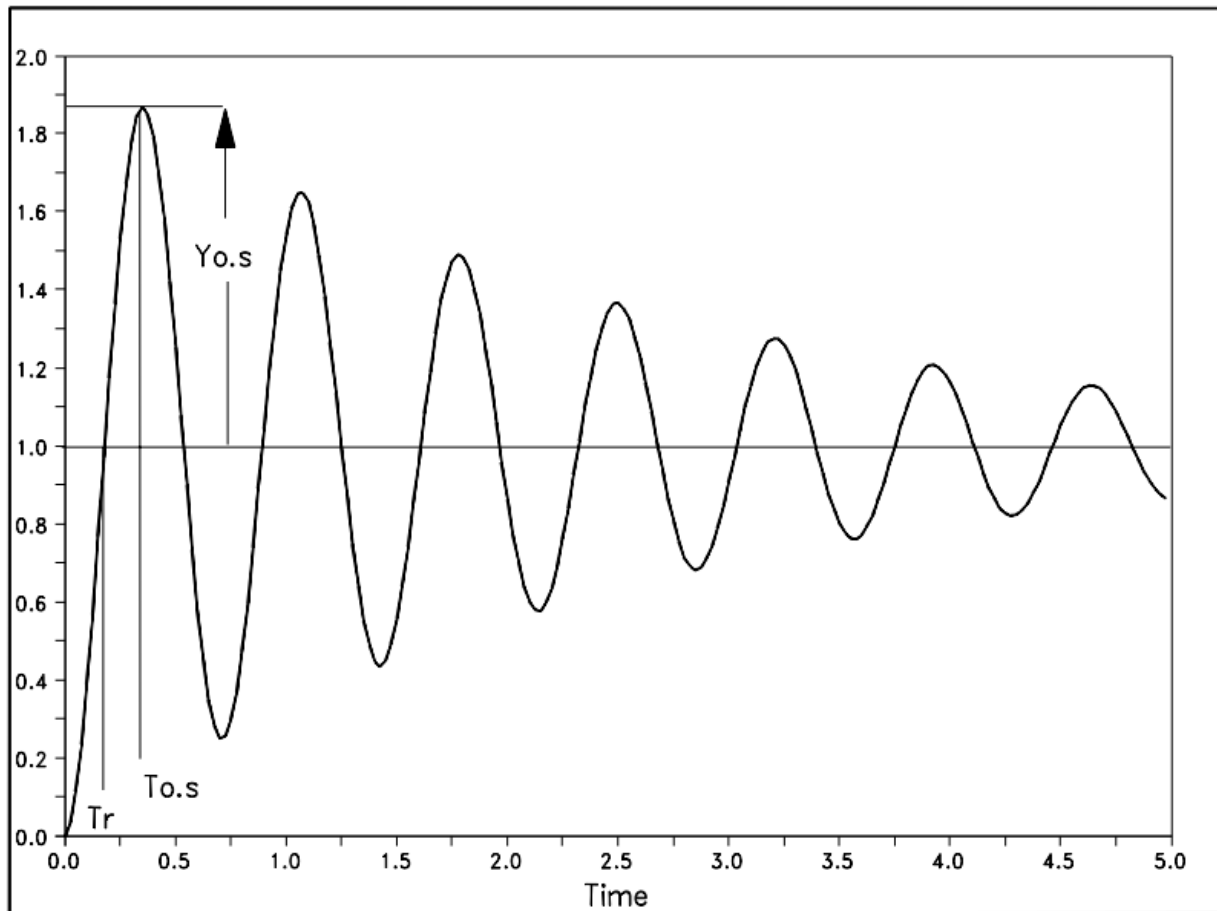


Fig. B. The step response of an under damped circuit.

II. INSTRUMENTATION

Function generator, Tektronix CFG 250. Oscilloscope, Tektronix 2205-20 MHz.

III. PROCEDURE

A. The Series RLC.

1. (a) Construct the series RLC circuit in Figure 1. Observe the capacitor voltage.
- (b) Set the signal of the source to the square pulse mode. Adjust the frequency of the signal to 10 times below the resonant frequency of the circuit.
- (c) Adjust R to 10 k Ω . What type of response do we obtain for the voltage?
- (d) Sketch on a millimeter paper the positive half cycle of the voltage wave form.
- (e) Adjust R to obtain the critical response. Record R.
- (f) Sketch on the same millimeter paper the positive half cycle of the voltage response.
- (g) Adjust R to 500 Ω . What type of response do we obtain for the voltage?
- (h) Sketch on the same millimeter paper the positive half cycle of the current response.
- (i) Measure the frequency, damping ratio, response time, and overshoot error.
- (j) Simulate the circuit of Figure 1 in PSpice. Use the same values of R as above to obtain the three types of response. Compare the simulated wave forms with the oscilloscope waveforms.

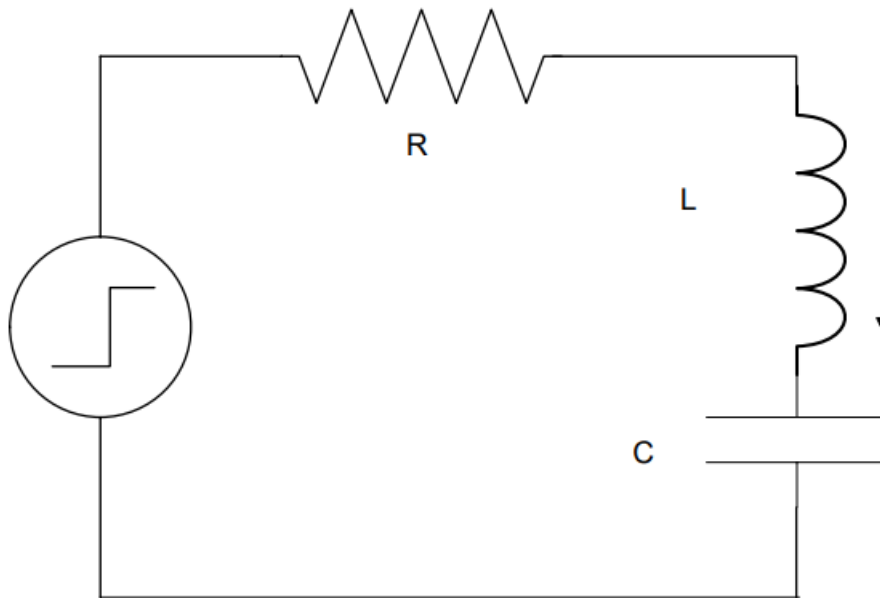


Fig. 1. The series RLC circuit. $C = 0.01 \mu\text{F}$, $L = 10 \text{ mH}$.

B. The parallel RLC.

2. (a) Simulate the circuit of Figure 2 in PSpice. Find the value of R resulting a critical response.
- (b) Set R to $2 \text{ k}\Omega$. Obtain the voltage response.

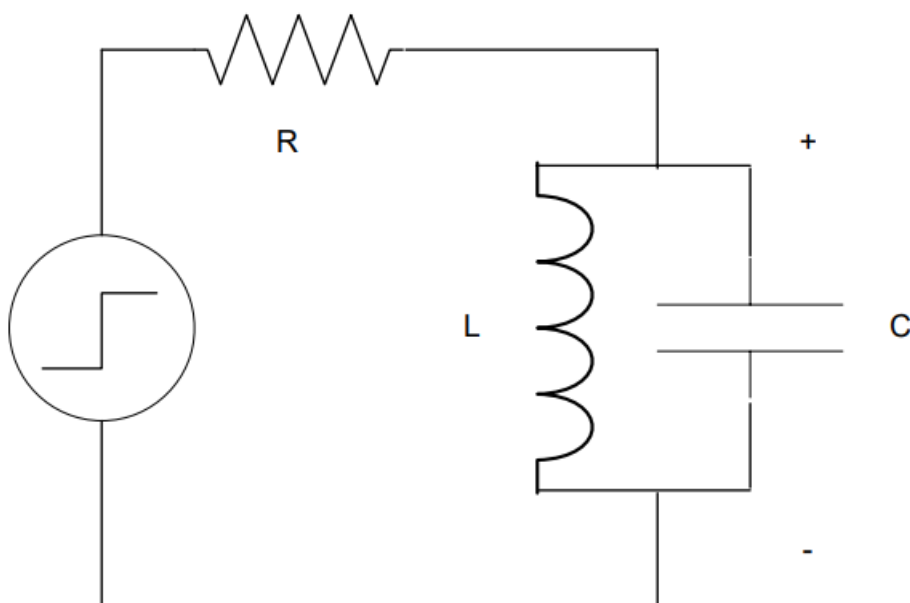


Fig. 2. The parallel RLC circuit. $C = 0.01 \mu\text{F}$, $L = 10 \text{ mH}$.

REPORT:

A. Theoretical Development.

1. In theory discuss the natural response of an RLC circuit.

B. Recording of the Three Types of Response of the Series RLC.

2. Are the three values of R in Procedures 1c, 1e, and 1g consistent with the theory?

3. Are the experimental responses consistent with those predicted by the theory? Explain by comparing the wave form parameters with their theoretical values in each case.

C. Under damped Response of the Series RLC Circuit.

4. Compare the oscilloscope wave form obtained in Procedure 1h with the PSpice simulation for the same circuit.

D. Simulation of the Parallel RLC circuit.

5. (a) Discuss the simulation results from Procedure 2a.

(b) From the simulation in Procedure 2b, calculate the maximum energy of the inductor and capacitor. Plot both as function of time. Plot their summation as function of time. Discuss the results.